MIL-C-52025C(ME)
6 December 1978
SUPERSEDING
MIL-C-52025B(ME)
15 May 1973

MILITARY SPECIFICATION

CONVERSION UNITS, CARBON-DIOXIDE, 260 POUNDS PER HOUR

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers two types of carbon dioxide conversion units having a rated capacity of 260 pounds per hour for converting liquid carbon dioxide to dry ice. The conversion unit may be housed and transported in two types of shelters. One, a semitrailer van and the other a 40-foot, International Organization for Standardization (ISO), commercial freight container body.

1.2 Classification. Conversion units shall be of the following types and sizes, as specified (see 6.2):

Type I - Semitrailer.
Type II - ISO container.

Size 40 - 1 ea. Container furnished.
Size 20 - 2 ea. Container furnished.

FSC 3655

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-F-475	- Floor Covering Linoleum.
W-C-375	- Circuit Breaker, Molded Case; Branch-Circuit and Service.
W-P-115	- Panel, Power Distribution.
BB-C-104	- Carbon Dioxide, Technical: Solid.
BB-F-1421	- Fluorocarbon, Refrigerants.
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
UU-T-81	- Tags, Shipping and Stock.
WW-C-624	- Coupling Assembly, Hose (Fire, Garden, Water Suction).
ZZ-H-451	- Hose, Fire, Woven-Jacketed, Rubber- or Cambric-lined, With Couplings.
ZZ-H-561	- Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water.
AAA-S-108	- Scale, Dial and Beam Indicating, (Bench 5 to 150 Pound Capacity).
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plate, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-T-740	- Truck, Tractor, 5 Ton, 6 x 6, Military Design, M39 Series.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

MIL-C-3432	- Cable and Wire, Electrical (Power and Control) Semi-Flexible, and Extra Flexible, 300 and 600 Volts.
MIL-R-3593	- Refrigeration and Cooling Equipment (Excluding Household Refrigerators), Packaging of.
MIL-C-3600	- Compressor, Rotary, Power-Driven; and Compressor, Reciprocating, Power-Driven: Air and Gas (Except Oxygen and Refrigerant), Packaging of.
MIL-W-5086	- Wire, Electric, Hookup and Interconnecting Polyvinyl Chloride-Insulated, Copper or Copper Alloy Conductor.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-S-12514	- Starters, Motor; Across-the-line and Reduced Voltage Types, A-C Induction, 3/4 to 100 Hp, Manual and Electric.
MIL-C-13489	- Cover and End Curtains; Cargo Body (For Military Vehicles).
MIL-P-14514	- Pump, Centrifugal, Self-priming, Frame-mounted, 1-1/2-Inch, 65 GPM at 50 Feet Total Head (For Water).
MIL-F-16081	- Fan, Ventilation, Propeller.
MIL-E-16298	- Electric Machines, Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-M-18058	- Machinery, Metal and Wood Working, Preparation for Delivery of.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-in.
MIL-C-22992	- Connector, Plugs and Receptacles, Electrical, Waterproof, Quick-Disconnect, Heavy Duty Type, General Specification for.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-S-40626	- Sign Kit, Vehicle Class.
MIL-T-40019	- Tank, Shipping and Storage, Dry Ice.
MIL-T-46786	- Tanks, Fuel, Engine: General Requirements for.
MIL-C-52286	- Cable Assemblies, Power, Electrical, (With Molded on Terminations).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED. STD. No. 66	- Steel: Chemicals Composition and Hardenability.
FED. STD. No. 595	- Colors.

MILITARY

MIL-STD-101	- Color Code for Pipelines and for Compressed Gas Cylinders.
MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging Eyes and Attachments for Lifting and Tying Down Military Equipment.
MIL-STD-281	- Automobile, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS33657	- Fitting End, Standard Dimensions for Bulkhead Flared Tube Connections.
MS35644	- Filler Neck Assembly, Fuel Tank-Military Vehicles.
MS35645	- Cap, Fuel Tank: Tactical Vehicle, Fording.
MS35844	- Plug, Machine Thread Magnetic; Drain.
MS49006	- Plugs, Pipe, Magnetic-Headless Aluminum, Iron, or Steel.
MS75021	- Connector, Receptacle, Electrical, 12 Contact, Intervehicular, 24 Volt, Waterproof.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:06Z
Check the source to verify that this is the current version before use.

(Copies of specifications and standards required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Section VIII - Boiler and Pressure Vessel Code, Unfired Pressure Vessels.
Section IX - Boiler and Pressure Vessel Code, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN PLYWOOD ASSOCIATION (APA)

US Product Standard PS 1.
Plywood Design Specification Manual X215.

(Application for copies should be addressed to the American Plywood Association, 1119 A Street, Tacoma, WA 98401.)

AMERICAN WELDING SOCIETY, INC. (AWS)

Standard Qualification Procedure.
Welding Handbook.
D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ISO Recommendation No. 668 - Dimensions and Ratings of Freight Containers.
ISO Recommendation No. 1496 - Specifications and Testing of Series 1 Freight Containers.
ANSI A119.2 - Mobile Vehicle Standards.
ANSI B31.1 - Code for Pressure Piping.
ANSI C1 - National Electrical Code.
ANSI MH5 - Specifications for Cargo Containers.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:06Z
Check the source to verify that this is the current version before use.

ANSI Z26.1 - Safety Glazing Materials for Glazing Motor Vehicles Operating on Land Highways.
ANSI Z35.1 - Industry Accident Prevention Signs, Specification for.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Steel Products Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

Book of ASTM Standards.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN BUREAU OF SHIPPING (ABS)

Guide for the Certification of Dry Cargo Containers.

(Application for copies should be addressed to the Secretary, American Bureau of Shipping, 45 Broad Street, New York, NY 10004.)

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Section VI - Rules Governing the Loading of Commodities on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 1920 L Street, NW, Washington, DC 20036.)

DEPARTMENT OF LABOR (DOL)

29CFR1910 - Occupational Safety and Health Administration Standards (OSHA).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF TRANSPORTATION (DOT)

49CFR393 - Federal Motor Carrier Safety Regulations (FMCSR).
49CFR420 - Transport International Routiers (TIR) Cornets.
49CFR450 - International Safe Containers.
49CFR567 - Certification.
49CFR571 - Federal Motor Vehicle Safety Standards (FMVSS).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

FORGING INDUSTRY ASSOCIATION

Forging Industry Handbook.

(Application for copies should be addressed to the Forging Industry Association, 1121 Illuminating Building, Cleveland, OH 44113.)

FRICTION MATERIALS STANDARDS INSTITUTE, INC (FMS)

Automotive Data Book.

(Application for copies should be addressed to the Friction Materials Standards Institute, Inc., East 210, Route 4, Parama, NJ 07652.)

INDUSTRIAL FASTENER INSTITUTE (IFI)

IFI Standards.
OMFS Recommendations.

(Application for copies should be addressed to Industrial Fastener Institute, 1505 East Ohio Building, Cleveland, OH 44114.)

NATIONAL ASSOCIATION OF METAL NAME PLATE MANUFACTURERS (NAME)

Metal Name Plate Industry Standards and Practices.

(Application for copies should be addressed to the National Association of Metal Name Plate Manufacturers, Inc., 1000 Vermont Avenue, NW, Washington, DC 20005.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA - STD.
MG1 - Electric Motors.
KS1 - Distribution Box.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 815 15th Street, Washington, DC 20005.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 - National Electrical Code (NEC).

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

REGULAR COMMON CARRIER CONFERENCE (RCCC)

RCCC Recommended Practices Manual.

(Application for copies should be addressed to the Regular Common Carrier Conference, American Trucking Association, Inc., 1616 P Street, NW, Washington, DC 20036.)

RUBBER MANUFACTURERS ASSOCIATION (RMA)

Engineering Standards for Multiple V-Belt Drives.

(Application for copies should be addressed to the Rubber Manufacturers Association, 444 Madison Avenue, New York, NY 10022.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013).

STATE OF CALIFORNIA DEPARTMENT OF MOTOR VEHICLES

California Vehicle Code.

(Application for copies should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

TIRE AND RIM ASSOCIATION, INC. (T&RA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

TRUCK-TRAILER MANUFACTURERS ASSOCIATION, INC. (TTMA)

TTMA Recommended Practices.
TTMA Technical Bulletins.

(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, Inc., 2430 Pennsylvania Ave., N.W., Washington, DC 20037.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

UNDERWRITERS LABORATORY INC. (UL)

UL Standards and Listings.

(Application for copies should be addressed to the Underwriters Laboratory Inc., 207 East Ohio Street, Chicago, IL 60611.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Description. The carbon dioxide conversion unit (hereinafter called "unit") shall be contained in a semitrailer van for the Type I, or one or two ISO freight containers, for Type II. The Type II unit shall be contained in one container for size 40 or two containers for size 20 (see 1.2 and 6.2). The operating components shall be housed within the enclosure, shall be electrically powered, and shall consist of a liquid carbon dioxide receiver, a carbon dioxide condensor, a carbon dioxide compressor, a monochlorodifluoromethane refrigeration system, a water pump, a dry-ice-press including an oil reservoir and pump, and all necessary manifolding hoses, instruments, gages, and electrical apparatus. A band saw and platform-scale shall be provided near the ice press. The unit shall convert 300 pounds of liquid carbon dioxide at 300 psig and 0° F to 260 pounds of dry ice per hour.

3.2 First article (preproduction model). The contractor shall furnish one or more units as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production, that his production methods and choice of design detail will produce conversion units that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified.

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more units as specified (see 6.2), for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.12).

3.4.1 Corrosion-resistant material. Where corrosion-resistant material is specified, corrosion-resistant steel conforming to FED. STD. No. 66, Type 301 through 348 as applicable, copper, brass, bronze, silver, beryllium-copper, copper-nickel alloys, or aluminum shall be used.

3.4.2 Corrosion-resistanc treatment. Corrosion-resisting treatment shall consist of galvanizing or electrodepositing of cadmium, chromium, copper or nickel.

3.4.3 Metallic parts. Sheet metal or metal stampings not more than 1/8 inch thick shall be nonferrous or corrosion-resistant steel. Zinc castings shall not be used. Latches, hinges, and hinge pins shall be either nonferrous metal, corrosion-resistant steel, or cadmium-plated steel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

3.4.4 Nonmetallic parts. Parts made of nonmetallic materials that could be adversely affected by normal contact with chemicals used in the production of carbon dioxide shall not be used in the construction of the conversion unit.

3.4.5 Semitrailer and body. Semitrailer and body materials shall be compatible with the intended use in accordance with the engineering standards of the van body, freight container, and semitrailer industry and as set forth in the publications of the Regular Common Carrier Conference, Society of Automotive Engineers and Truck-Trailer Manufacturers Association. Unless otherwise specified herein, rubber products furnished shall not exceed 2 years in age from date of cure of rubber to date of acceptance of the unit by the Government.

3.5 Environmental conditions.

3.5.1 Operating temperature requirements. The unit shall perform as specified herein in any ambient temperature from plus 125° F or 51.67° C to minus 25° F or minus 31.67° C.

3.5.2 Storage temperature requirements. The unit shall not be damaged by storage at temperatures (after stabilization) from minus 50° F (minus 45.6° C) for 6 hours with no solar radiation, to plus 155° F or plus 68.3° C) for 1 hour with no solar radiation.

3.5.3 Precipitation. The unit shall not leak water into the interior of the body when exposed to rain at a rate of 3 inches per hour impinging at angles varying from the vertical to 45 degrees when directed simultaneously at the roof and each of the four sides. Operation of the unit shall not be affected by the precipitation nor shall operators be exposed to the precipitation.

3.6 Plant safety. The unit shall comply with DOL OSHA and DOT requirements for hazardous equipment.

3.6.1 Physical, mechanical, and electrical hazards. All belts, gears, shafts, pulleys, sprockets, spindles, drums, flywheels, chains, and other reciprocating, rotating, or moving parts of equipment shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of equipment, including exhaust pipes and other lines which may be subject to high temperatures, exposed to contact by personnel or which create a fire hazard shall be fully guarded or insulated. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto engine, exhaust, or electrical equipment. Exhaust or discharges from equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. All points requiring lubrication during

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

operation shall have fittings located or guarded so as to be accessible without hazardous exposure to personnel performing these operations. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided.

3.6.2 Fire and explosion hazards. Construction of unit components shall be such as to eliminate fire and explosion hazards.

3.6.3 Pressure relief devices. Relief valves and rupture disks shall conform to ASME Boiler and Pressure Vessel Code, Section VIII. Relief valves shall be provided for relief of pressures in excess of normal operating pressures. The discharge from all pressure release valves (liquid or gas) shall be directed or manifolded to a safe location so as to not endanger operating or maintenance personnel. All safety valves shall be connected to a manifold and vented to the outside of the van.

3.7 Vehicle safety. The safety characteristics of the semitrailer portion of the unit shall meet the following minimum requirements:

(a) Eliminate concealed danger created by the contractor's design.
(b) Eliminate hazards to personnel and to property.
(c) Result in furnishing material having a correct safety factor.
(d) Provide safety and protective devices which do not impair the operating functional requirements or furnish prominent warnings.
(e) Conform to all requirements pertaining to large vehicles of the DOT Federal Motor Vehicle Safety Standards (FMVSS) and Federal Motor Carrier Safety Regulations (FMCSR), except as specified herein.
(f) Conform to all requirements of the California Vehicle Code Division 12, except as specified herein.
(g) All walk and step surfaces shall provide a nonskid footing in accordance with SAE J185. Nonfunctional sharp edges, projecting points, and excessive length of fasteners shall be eliminated. Human factors engineering characteristics shall be incorporated into the plant.

3.8 Human factors engineering. The characteristics of the unit shall provide for operation by personnel ranging from the small man, clothed, through the large man, arctic-clothed in accordance with SAE J833. The minimum access dimensions shall conform to SAE J925 for construction and industrial machinery.

3.8.1 Sound level. The sound level of the unit shall be in accordance with DOL OSHA and with MIL-STD-1474, Category D, for steady-state noise

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

3.9 Reliability. The specified Mean-Time-Between-Failure (MTBF) shall be 115 hours when tested as specified in 4.6.2.5.

3.10 Maintainability.

3.10.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts, as specified (see 6.8 and 6.11), shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.6.2.4.

3.10.2 Maintenance ratio. The units shall have a maintenance ratio of not more than 0.06. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.10.3 Scheduled maintenance. The units shall not require scheduled maintenance more often than every 8 hours of operation when tested as specified in 4.6.2.4.

3.10.4 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners. All exposed fasteners shall be of corrosion-resistant material or shall be treated to be corrosion resistant (see 3.4.2). Means for complete drainage of water, lubricants, fuel, and operating liquids in all components and piping shall be provided in an accessible location and shall drain to the ground without splashing on any component. Draining manifolding shall be of sufficient capacity to completely drain any reservoir within 20 minutes after opening drain cocks.

3.11 Performance.

3.11.1 Production capacity. The unit shall produce 260 pounds of dry ice per hour, in block form or, when specified (see 6.2), in pellet form. The product shall conform to BB-C-104. The unit shall perform at the specified capacity under all environmental conditions specified herein. The unit shall perform as specified herein when utilizing not more than 65 gallons of cooling water per minute at temperatures up to plus 90° F or plus 32.22° C.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:06Z
Check the source to verify that this is the current version before use.

3.11.1.1 Start up. Full production capacity and product quality shall be attained within 1 hour when the unit is started with all unit components at the operating conditions specified in 3.11.1.

3.11.1.2 Endurance. The unit shall operate for not less than two, 125 hour production periods when tested as specified in 4.6.2.6. During each 100 hour endurance test period, a shutdown not to exceed 60 minutes duration shall be permitted once each 24 hours for performing necessary preventive maintenance, lubrication, and minor repairs such as adjusting press and packing glands. This hour shall not be counted as test time. Upon resuming operation after the shutdown period, the unit shall produce the product at the specified capacity within 30 minutes after startup.

3.11.1.3 Shutdown and preparation for movement. Shutdown of the unit, from full production, and preparation for movement to a new location, by transporting on the highway and off-road travel, shall be accomplished within 2 hours by two operators.

3.11.1.4 Setup after movement. Setup and preparation of the unit for operation shall be accomplished within 2 hours by two operators, after the unit has been transported to a new location.

3.11.2 Transportability.

3.11.2.1 Mobility characteristics. The Type I unit shall conform to the mobility characteristics specified herein without malfunction, permanent deformation, or damage when: (1) Prepared for transport; and (2) when towed by a Government 5-ton 6 x 6, as shown on MIL-T-740 or equal commercial truck tractor. The Type II unit shall also operate as specified herein when securely mounted on an ISO container semitrailer or low bed semitrailer of sufficient length to accommodate the container.

3.11.2.2 Highway travel. The unit shall travel over smooth, hard-surface primary highways at sustained speeds up to 60 miles per hour (mph).

3.11.2.3 Secondary road travel. The unit shall travel over gravel-surfaced secondary roads at sustained speeds up to 30 mph.

3.11.2.4 Off-road travel. The unit shall travel over unimproved roads, open fields, rolling hills, and rough off-road terrain at speeds up to 10 mph. The unit shall negotiate grades up to 30 percent, traverse side slopes up to 10 percent, and shall travel through 9 inches of mud, sand, and snow to the intended operation sites. Without prior preparation the unit shall ford hard-bottom water crossings up to 20 inches deep without damage or adverse effects due to immersion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

3.11.2.5 Trailing and turning ability. The unit semitrailer shall trail the truck-tractor without bottoming, cramping, weaving, or sidesway. The unit shall follow the truck-tractor without exceeding the tracking deviation, as specified in DOT FMCSR, 393.70 when towed over a paved surface. Jackknife turns of at least 90 degrees, as measured from straight ahead to the right and to the left, shall not cause damage to or interference between the truck-tractor and the unit. The unit shall not be damaged by pivot turns about the center of the unit suspension.

3.11.2.6 Braking ability. The combined service brakes of the truck-tractor and the unit shall control, decelerate, and stop the vehicle combination, all in accordance with DOT FMVSS, DOT FMCSR, and the California Vehicle Code requirements. The brake system of the unit shall be in accordance with TTMA RP No. 12 and, if legally required, RP No. 44.

3.11.2.7 Rail transport. The unit shall be rail transportable when loaded on a flatcar in accordance with the Association of American Railroad's "Rules Governing the Loading of Commodities on Open-Top Cars". The unit shall withstand all forces encountered in piggy-back rail transport services (TOFC) without breakage or permanent deformation. For purposes of rail transport, the unit shall be compatible with trailer-train rail cars.

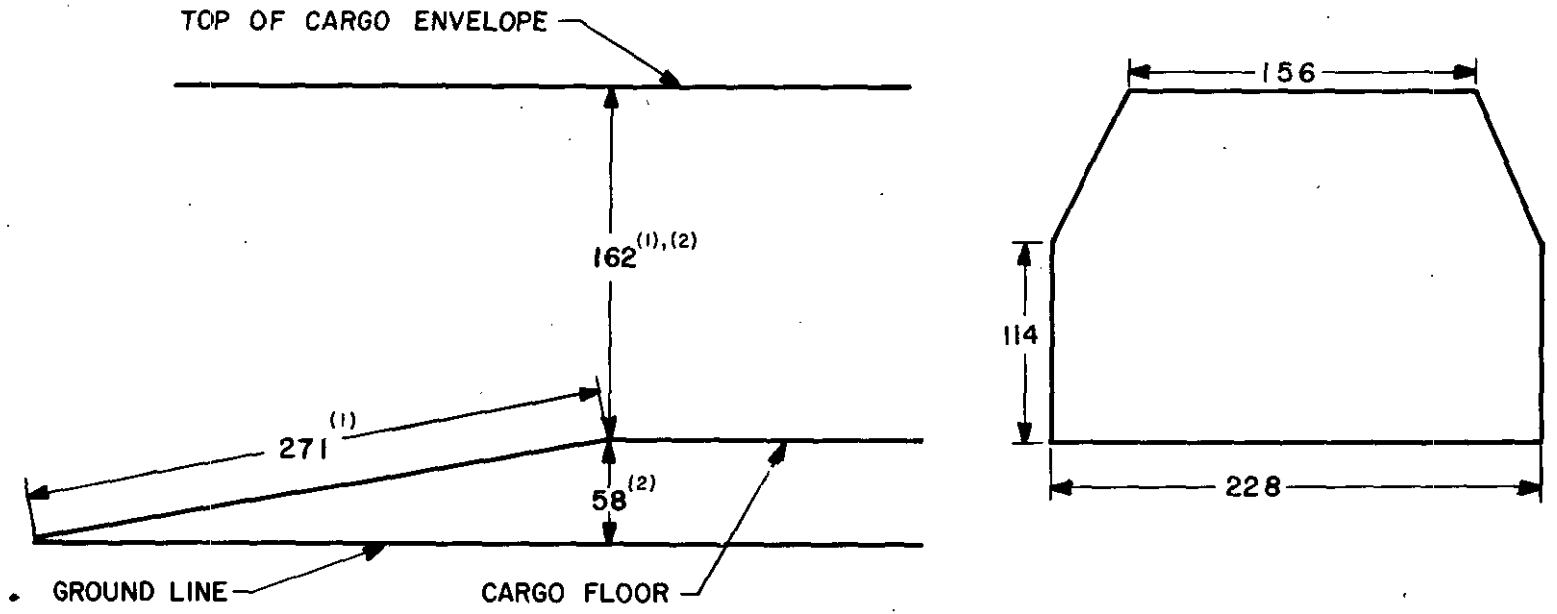
3.11.2.8 Air transport. The unit shall be transportable when loaded in C5A aircraft as shown in Figure 1.

3.11.2.9 Marine transport. The unit shall be transportable on Roll-on/Roll-off ships and, for Type II units, on board container ships.

3.12 Construction. The unit shall be constructed so that no parts will work loose in service. The unit shall be built to withstand without damage the strains, jars, vibrations, and other detrimental conditions incidental to normal operation, maintenance, shipping, and storage. Unit components which do not have to be removed or repositioned for operation shall be securely mounted to the unit frame floor or to other major components. All subbases and other fixtures required for secure component mounting shall be provided. Unit components shall be located and installed to insure accessibility of controls and instruments to the operator when he is working at the control area, and to provide for the minimum lengths of lines and tubing between components. The components shall be located and installed within the unit body to insure equal weight distribution and tire loadings during transport.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

FORWARD CARGO OPENING C-5A
FORWARD KNEEL POSITION



NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. REFERENCE: AMCP 706-130, ENGINEERING DESIGN HANDBOOK, DECEMBER 1967.
3. REFERENCE: USATECOM PROJECT NO 4-ES-065-050-004, SERVICE TEST OF C-5A AIRCRAFT, FIRST PARTIAL REPORT BY H.H JOHNSON, JR., CPT., JANUARY 1971.

FIGURE 1. C-5A. cargo envelope.

X-3171A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

3.12.1 Fastening devices. All screws, pins, fastener assemblies, and other similar devices shall be installed with a positive means of protection against working loose due to vibration. Locking shall be by compound, cotter pins, prevailing torque locknuts, castellated nuts, or any other approved locking method. Swedging, peening, welding, or staking of threads of parts which are subject to removal or adjustment will not be permitted. When tightened in place, threaded fasteners shall have approximately two threads showing. Screw threads shall be in accordance with IFI Standards and SAE Standards.

3.12.2 Piping fittings. Piping and fittings connecting processing elements shall be installed so that the piping will not interfere in any manner with accessibility to components of the unit. Unions or flanges shall be located in the pipe to permit removal of connected components for repair or replacement. Piping shall conform to ANSI B31.1 and shall be braced and supported to minimize vibration or racking. In no case will functional piping be used as a support for other piping or equipment. All tubing, and electrical conduit shall be located in protected positions, shall be securely fastened to frame or body structures, and shall be furnished with metal protective loops or grommets at each point where they pass through members, except where a through-frame connector is provided. Provisions shall be made for connecting all service hose including water pump discharge, cooling water intake, cooling water discharge, liquid CO_2 intake, and the CO_2 equalizing line on the exterior wall on the curb side of the unit. All connections shall be clustered on a metal plate attached to the unit wall. The cooling water inlet shall be 1-1/2 inch NPT male union connection, the water pump discharge and cooling water discharge connections shall be 1-1/2 inch NPSH male connections. The liquid CO_2 intake and the CO_2 equalizing line connections shall conform to MS33657. A thread protecting cap, attached by chain, shall be provided for each connection.

3.12.3 Pressure vessels. Unless otherwise specified herein, all pressure vessels shall be fabricated with not less than a 4 to 1 safety factor based on the minimum tensile strength of the material, as specified in the ASME Boiler and Pressure Vessel Code, Section VIII. Presence of the ASME official code U-symbol stamped or marked on the vessel and the furnishing of a copy of the ASME data sheet (Form U-1) may be accepted as evidence that the vessel conforms to the ASME Rules for Construction of Unfired Pressure Vessels.

3.12.4 Pressure relief devices. Relief valves or rupture disks shall be provided to protect against excessive pressure in any component. Relief valves and rupture disks shall conform to ASME Boiler and Pressure Vessel Code, Section VIII. Relief valves shall be provided for relief of pressures in excess of normal operating pressures. Rupture disks may be provided

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

where pressure will not normally have to be relieved, but where protection is necessary. The discharge from all pressure release valves (liquid or gas) shall be directed or manifolded to a safe location so as to not endanger operating or maintenance personnel.

3.12.5 Liquid-level gages. Liquid-level gages shall be provided on all vessels containing fluids and shall be for the service intended. Liquid-level gages provided for use on cold vessels shall remain frost free and readable when in use. All liquid-level gages shall be shielded to prevent explosion and shall be provided with manual shutoff valves. Relief valves shall be provided to release pressure buildup from liquid trapped in the liquid level gage.

3.12.6 Pressure and temperature indicators. Pressure and temperature indicators shall be provided for each component of the unit, where constant monitoring of pressure and temperatures are essential to effective and safe operation. All indicators shall be positioned so as to be visible to the operator from the dry-ice press control area.

3.12.7 Drain connections. Drain connections and cocks shall be provided throughout the unit on all components and connecting piping. Piping and components of the unit shall be designed and located for convenient and positive self draining. All drain and blowdown lines shall be extended through the van floor.

3.13 Components.

3.13.1 Carbon dioxide receiver. The carbon dioxide receiver shall be an insulated pressure vessel of sufficient size to provide liquid carbon dioxide to produce dry ice as specified herein. The receiver shall be provided with a liquid-level gage, pressure gage, vent, safety valve, and rupture disk. The receiver shall be serviced with 3/4 inch flexible fill and pressure equalizing lines.

3.13.2 Dry-ice press. The dry ice press shall be a minimum 62-ton, portable, universal unit of the 3-cylinder type. The press shall produce not less then 260 pounds of solid carbon dioxide, (dry ice) conforming to, BB-C-104, per hour. Dry ice blocks shall be 10 inches by 10 inches by 5 or 6 inches in thickness. The press shall receive liquid carbon dioxide at 300 psig and 0° F. The height of the press shall be not more than 7 feet 11 inches with the compression ram in top position, and the weight shall be not more than 7,000 pounds. The press shall be complete with all necessary accessories, such as 4-way control valves, oil reservoir, pumps, interconnecting piping, pressure gages, safety valves, thermometers, timer, signal

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

light, and alarm bell. The press shall be provided with a 1,000-watt electrical, resistance-type heater attached to the control needle valve of the feed line of the press to thaw the needle valve. Unless otherwise specified (see 6.2), a 1,000-watt, resistance-type heater shall be provided to thaw water-ice and to prevent its formation on the press ring seal. A seal permanently bonded to the lower platon shall be provided to prevent leakage of CO_2 while pressing the dry ice into cake form.

3.13.2.1 Press oil reservoir. The press oil reservoir shall have a 50-gallon capacity. The reservoir shall be provided with a drain and drain control located at the lowest point. The oil sump of the reservoir shall be provided with three 1,000-watt, 3-heat, immersion, bayonet-type heaters and a water cooling coil. The heaters shall be capable of maintaining the temperature of the oil at not less than 70° F in the installed position when the unit is subjected to the lowest ambient temperature specified in 3.5. When supplied with water at plus 90° F, the water cooling coil shall maintain the oil at a temperature not greater than plus 110° F when the plant is subjected to the highest ambient temperature specified in 3.5. The water cooling coil shall be rigidly installed, capable of complete drainage, and located so as not to interfere with the other lines and piping in the oil reservoir.

3.13.2.2 Double, 4-way operating valve. A double, 4-way operating valve, having a 2-position, 2-lever valve with internally built-in check valve to hold sealing pressure on the ram during reversal of the valve piston, shall be furnished. Two single 4-way valves may be furnished in lieu of a double 4-way valve when available space renders the double valve impractical. The 4-way valve shall be covered with sufficient insulation to prevent any accumulation of frost or ice formation during operation or standing.

3.13.3 Pressure regulators. The conversion unit shall be provided with a pressure regulator on the low-pressure side in the line between the dry ice press and carbon dioxide compressor. The regulator shall operate on carbon dioxide gas supplied within a pressure range of 70 to 76 psig and shall maintain a downstream pressure at any setting in the range of 20 to 50 psig within 1 psig, with a capacity of 300 pounds per hours of carbon dioxide. In addition, the conversion unit shall be provided with a pressure regulator on the high-pressure side in the line between the compressor discharge and the compressor suction line. The regulator shall maintain the downstream pressure within the range of 5 to 7 psig and the upstream pressure is in the range of 280 to 320 psig. The regulators shall have a capacity of 300 pounds of carbon dioxide per hour.

3.13.4 Carbon-dioxide compressor. The compressor shall operate at a suction pressure from 5 to 35 psig. The compressor shall have a capacity of 360 pounds per hour of carbon-dioxide gas at a suction pressure of 30 psig.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

The compressor shall have a forced-feed, lubricating-oil system which shall have a low-oil pressure safety device and high-oil-pressure relief valve. The lubricating oil system shall maintain a positive pressure of not less than 35 psig in normal operation. The compressor shall be provided with anti-friction bearings and porous, chrome-plated cylinder liners or chrome-plated cylinders. The compressor intercooler shall be shell-and-tube unit, with fixed tube sheets. The operating pressures shall be not greater than 100 psig on the shell side. The intercooler shall handle carbon-dioxide gas on the shell side and cooling water on the tube side. The compressor aftercooler shall be a shell-and-tube unit, with fixed tube sheets. The operating pressure shall be not greater than 350 psig on the shell side. The aftercooler shall handle carbon-dioxide gas on the shell side and water on the tube side, with a pressure drop not greater than 5 psig on the shell side and 3 psig on the tube side. A surge pot shall be provided in the compressor suction line.

3.13.5 Carbon-dioxide heat interchanger. A carbon dioxide heat interchanger shall be provided and shall pass the return gas from the dry-ice press against the gas coming from the compressor aftercooler prior to its introduction into the condenser. The heat interchanger shall be a shell-and-tube type. The shell side shall have an operating pressure of 350 psig and the tube side, 35 to 50 psig. The pressure drop shall be not more than 5 psig on the shell side and not more than 2 psig on the tube side.

3.13.6 Refrigeration system. The main components of the refrigeration system shall be a refrigerant compressor, an electric motor, a condenser, an oil separator, a refrigerant receiver, a heat exchanger and a refrigerant drier. All components shall be mounted on a common base. A float type syphoning tube shall be provided in the refrigerant receiver to assure that the compressor oil carrier over with refrigerant gasses will be returned to the compressor crankcase as it accumulates. The refrigeration system shall be of sufficient size and capacity to provide the refrigeration necessary to meet the requirements of the conversion unit when operated as specified herein. The refrigerant used shall be halogenated hydrocarbon conforming to BB-F-1421, Type 22. The refrigeration system manifolding shall conform to the requirements of ANSI B31.1.

3.13.7 Carbon-dioxide condenser. The condenser shall be of the horizontal type with removable head and tube bundle. The condenser shall have a capacity of 400 pounds of carbon dioxide per hour at a working pressure of 300 psig.

3.13.8 Extrusion type pellet machine. When specified (see 6.2), an extrusion type pellet machine, of the reciprocating or rotary design shall be supplied in lieu of the dry-ice press and its components, the bandsaw,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

and weighing scale. The pellet machine shall be of the closed type, recovering all flash-off gas and returning it to the liquid receiver. The machine shall have a variable production capacity of, 0 through 500 pounds of pellets per hour after cooldown. The maximum cooldown period time shall not exceed 30 minutes. The pellet sizes shall range from 1/8 inch through 3/4 inch in diameter. Pellet sizes shall be in increments of 1/8 inch. The pellets which are laminated extrusions shall not exceed 1 inch in length. Provisions shall be made for bagging and sealing the pellets for shipping. Unused floor space realized from the removal of the dry-ice press and other equipment shall be utilized when specified (see 6.2), for housing a larger capacity liquid CO_2 receiver, and the pellet bagging and storage operation. The power requirement for the pellet machine shall not exceed that required for the ice press.

3.14 Accessories.

3.14.1 Scale. A scale conforming to AAA-S-108, Type I, dial range 75 pounds, shall be provided. The scale shall be equipped with a locking device for locking while in transit. The scale shall be securely mounted on a frame located near and at the level of the ice press and the bandsaw table.

3.14.2 Bandsaw. The conversion unit shall be provided with a bandsaw for cutting the solid carbon-dioxide blocks of the dimensions specified. The saw shall have a speed capacity sufficient to cut the blocks into 1-inch cubes as they are produced while the unit is operating at the capacity specified. The bandsaw shall be securely mounted in the area of and at the level of the scale.

3.14.3 Ventilation fan. A ventilation fan shall be provided near the floor of the unit immediately under the bandsaw, to remove carbon dioxide vapors from the operations area. The fan and motor shall conform to MIL-F-16081, Type I, Class 1, Size 16, Style A. The wall opening shall be fitted with weighted louvers that will be fully opened by the air displaced while the fan is in operation. The louvers shall automatically drop to the closed position when the fan is not in use. The louvers shall be weather and watertight when closed. Provisions shall be made for locking the louver opening closure from inside the conversion unit.

3.14.4 Water pump. A water pump shall be provided to supply cooling water for the refrigeration condenser, and the carbon dioxide compressor. The pump shall conform to MIL-P-14514, Type I, except that the motor shall be 416-440 volt maximum, 3-phase, 60 cycle. The pump shall be of the portable type, for use inside or outside the unit. Provisions shall be made for securing the pump base to the unit floor. The inlet and outlet

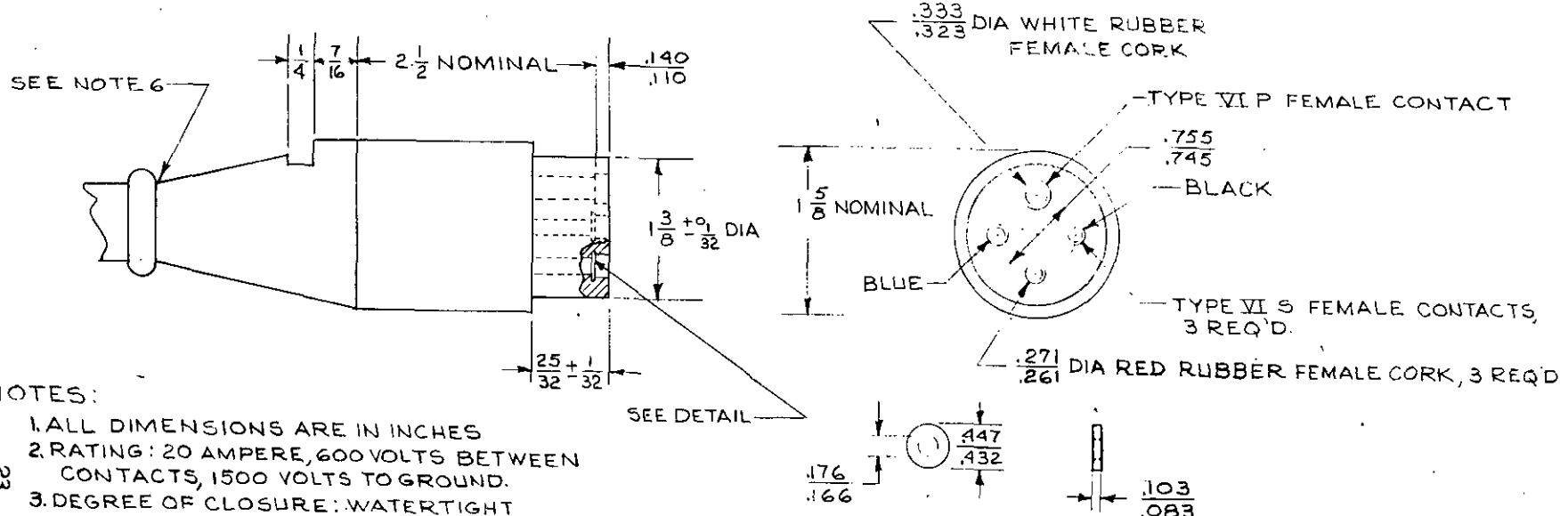
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

connections inside the unit shall be compatible with those specified in 3.14.5. Provisions shall be made outside the unit body for a water inlet when the pump is used outside the unit. A 50-foot rubber insulated cable of suitable size to power the water-pump motor under full load shall be furnished. One end shall be connected to the pump motor junction box and the other end shall be fitted with a female receptacle as shown in Figure 2. A male power supply receptacle shall be provided on the main panelboard. The connector receptacle shall have a metallic shell with a spring-loaded door which will close automatically when the connector plug is withdrawn. The connector receptacle shall encase the male connector plug as shown in Figure 3. A clearly marked storage area with securing facilities shall be provided on the pump for storing the cable.

3.14.5 Water-pump hose. A 1-1/2 inch discharge hose 50 feet long conforming to ZZ-H-451 shall be furnished. The hose shall be fitted with a 1-1/2-inch male NPT on the pump and 1-1/2-inch NPSH female connection on the unit end. Two 1-1/2-inch suction hoses 20 feet long conforming to ZZ-H-561, Type I, shall be furnished. The hose shall be supplied with a strainer and a foot valve on one end and a 1-1/2-inch male NPT on the pump end. A clearly marked storage area with securing facilities shall be provided omside the unit for the water-pump hoses when they are not in use. The couplings shall conform to WW-C-624. Adapters shall be provided to connect the hose to the equipment.

3.14.6 Liquid carbon dioxide transfer and equalizer hose. Two 50 foot, 3/4 inch, flexible carbon dioxide hoses suitable for handling pressures up to 600 psig, at temperatures to minus 65° F shall be furnished. All fittings and adaptors shall be furnished for attaching each hose to the unit as specified in 3.12.2 and at the supply source. Protective covers shall be provided at each end of the hose. The protective covers shall be permanently attached to the hose by swivel chains.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.



NOTES:

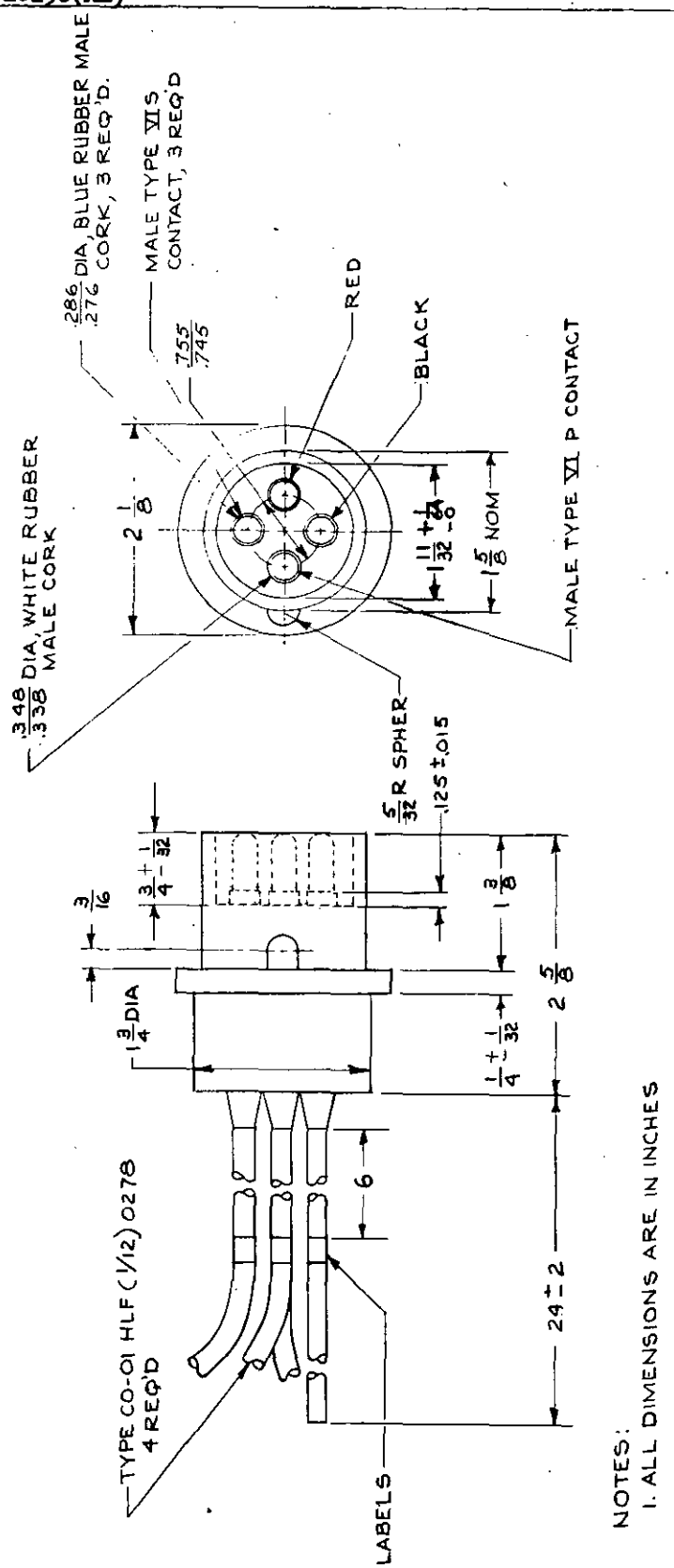
1. ALL DIMENSIONS ARE IN INCHES
2. RATING: 20 AMPERE, 600 VOLTS BETWEEN CONTACTS, 1500 VOLTS TO GROUND.
3. DEGREE OF CLOSURE: WATERTIGHT
4. CONNECTING AND DISCONNECTING FORCE: 10-35 POUNDS
5. INSULATION SHALL BE TYPE IS-L AND JACKET SHALL BE TYPE JS-L PER MIL-I-3930
6. DIMENSIONS OF GROOVE TO BE SUFFICIENT TO RETAIN A 1/8 DIA CORD OR TWINE TIED ON TO THE CONNECTOR; SHAPE AND POSITION NEED NOT BE AS SHOWN.

FIGURE 2.

CONNECTOR, PLUG, ELECTRICAL, FEMALE

X-2454

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.



NOTES:
1. ALL DIMENSIONS ARE IN INCHES

FIGURE 3.
INSERT, ELECTRICAL, CONNECTOR MALE

X-2455

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

3.14.7 Space heaters. Two fuel oil burning heaters, complete with all accessories, controls, and ducts shall be furnished for heating and maintaining the temperature inside the unit at not less than plus 50° F or plus 10° C when operating in the lowest ambient temperature specified in 3.5. The heaters shall be shock mounted and shall be of the centrifugal-pressure type with separate, electric motor driven blowers for supplying fresh air from outside the unit to the burner for combustion and to the exterior surfaces of the combustion chamber and heat exchanger for heating the air entering the duct system. Each heater shall be connected to an individual ducting system. All ducts shall be fitted with air grills or registers having manual, 4-way air direction adjustment. Provision shall be made for the use of 100 percent fresh air, 100 percent circulated, recirculated air, or any proportion of each desired by the operator. Weatherproof and insect-screened fresh air intakes shall be furnished in each side of the unit. A 110-volt ac, electric-motor-driven fuel pump shall be furnished. The fuel pumps shall be positioned in a protected location near the fuel tanks. The pump suction line shall not exceed a vertical lift distance of 24 inches above the top of the tank. Fuel lines shall conform to SAE J1402, Type D, Class II with reusable swivel fittings. Fuel overflow lines from each heater shall extend below the units. The heater exhaust lines shall extend either sideward or forward through the walls of the unit. The heater exhaust lines shall be of corrosion-resistant steel. The exhaust lines shall have no pockets which cannot be cleaned or drained. All exhaust lines shall be insulated or protected to preclude damage to seals, gaskets, and wiring and to prevent deformation caused by heat. The heating system shall conform to the requirements in ANSI A119.2 with respect to plumbing, heating, health, and safety features.

3.14.7.1 Fuel tank. A heater fuel tank conforming to MIL-T-46786, Type I, Class 1, shall be provided and shall have a minimum fuel capacity of 25 gallons. The filler-neck assembly and the cap shall be in accordance with MS35644 and MS35645, respectively. A strainer shall be provided in the filler-neck assembly. The fuel tank service outlet shall come out the top of the tank. The fuel service outlet shall be not less than 1/2-inch above the bottom of the tank. A fuel level gage shall be provided in the tank. The fuel tank shall be mounted either below the floor or in an interior vapor-tight enclosure with outside vent with the recessed fill connection at the side of the unit.

3.15 Utility electrical power system. The operating machinery of the conversion unit shall perform as specified herein when powered by an external electrical power source of 416 or 480 volt, 3 phase, 60 Hertz, alternating current (ac). Electrical components shall be in accordance with NEMA Standards and UL listings. The utility electrical power system shall conform to and be installed in accordance with the National Electrical Code

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

(NEC), ANSI C1 or National Fire Protective Association (NFPA) Pamphlet No. 70. The electrical power system shall distribute power for operation of machinery, equipment, lights, and heaters.

3.15.1 Main power supply. Unless otherwise specified (see 6.2), the following cables, plugs, and receptacles shall be provided. The main power cable shall conform to MIL-C-3432, Type 44-13-CO-04 HDF (4/1 - 4/8R) 1620. The overall length of the power cable shall be not less than 100-feet and if the weight of the cable assembly exceeds 200-pounds, the power cable shall be furnished in two lengths of at least 50-feet each. The live, input end of the main power supply cable shall be equipped with bare conductors, that are exposed for connecting to the solderless terminals on an external utility power source. The other end of the cable assembly, that feeds into the van body, shall be equipped with a standardized, polarized plug in accordance with MIL-C-22992, Class L. The plug and its mating receptacle shall be rated for the maximum current demand of the conversion unit equipment and shall be not smaller than size shell No. 44, 3 phase, ac, 4 wire, grounding, having shell marker No. 5 (135°) for 210/416 VAC or shell marker No. 6 (150°) for 277/480 VAC. The plug shall include a captive fastened, protective cover. The van body shall be equipped with a main power supply receptacle and captive cover in accordance with MIL-C-22992, Class L, 4 wire, 3 phase, ac, 60 Hertz. The receptacle shall mate with the cable plug. The main power receptacle shall be recessed in the curbside wall of the van body, near the lower rear corner. The receptacle cover shall be attached to the van body by means of a swivel captive fastener. The receptacle housing shall be connected to the system ground conductor and to the ground pin. The lead-in cable from the receptacle to the distribution box shall be directly connected to a double-throw switch as specified. The lead-in cable shall be encased in rigid conduit.

3.15.2 Auxiliary power supply. When specified (see 6.2), an auxiliary power cable conforming to MIL-C-3432, Type CO-03HLF (3/12) 0655 shall be furnished. The overall length of the power cable shall be not less than 100 feet. The load end of the cable shall be provided with a 3-pole, molded-rubber, socket-plug connector conforming to MIL-C-52286 and to Figure 1. The other end of the cable shall be bare conductors for connection to the solderless terminals on a separate generator set (outside source). Means shall be provided for storing the cable when it is not in use.

3.15.2.1 Auxiliary power supply receptacle. A weatherproof pin receptacle connector, which mates with the socket-plug connector specified in 3.15.2, shall be furnished. The auxiliary power receptacle shall be recessed next to the main power receptacle. The cover shall be spring loaded. The lead-in cable from the receptacle shall be connected directly to a double-throw switch as specified. The lead-in cable shall be encased in rigid conduit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

3.15.3 Electric controls. Electric controls, switches, protective devices, motor starter, transformers, circuit breakers, and light switches shall be grouped into a master control center. The master control center shall be installed in a location convenient to the operator when working at the dry-ice press control valves. All electric controls shall be installed for use with 120-volt, single-phase, 60-cycle, ac operating power from the main power supply system through a transformer or from the auxiliary power supply system.

3.15.4 Panelboards. Panelboards shall be of the circuit-breaker type. Panelboards shall conform to W-P-115, Type I, Class 1. Panelboards shall be of the dead-front safety type. Circuit breakers shall conform to W-C-375 and shall be of the ground fault circuit interrupter (GFI) type. Circuit breakers shall be of the bolt-in type with screw terminals including lockwashers and nuts. A solid insulated groundable neutral shall be provided within the enclosure. Trip elements shall be noninterchangeable. Wiring terminals shall be provided with pressure wire connectors; plug-in terminals shall not be used. At an ambient temperature of minus 25° F, a circuit breaker carrying 150 percent of its rated current shall operate (break automatically) within 2 hours.

3.15.4.1 Main panelboard. The main panelboard shall consist of a main circuit breaker to apply and to remove power from the distribution circuits, and a distribution panel. The main panelboard shall consist either of two units, one for the main circuit breaker and one for the other circuit breakers, or a single unit containing both main and other circuit breakers.

3.15.4.2 Auxiliary panelboard. The auxiliary panelboard shall provide circuit breakers as indicated below:

(a) Maintenance tool circuit.
(b) Plant-lighting, water pumps, air compressor, sump pumps, signal lights, and alarm bell.
(c) Ventilation fan.
(d) 120-volt interior lighting circuit.
(e) Alternate fuel oil heater circuit.
(f) Dry-ice press needle valve heater.
(g) Dry-ice press, press ring heater.
(h) One spare.

3.15.4.3 120 volt power supply switch. A double-throw switch shall be provided in accordance with NEMA Standard KS1, Type I enclosure. The switch center terminals shall be connected to the distribution panel with alternate supply from either the main power supply or from the auxiliary power supply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

3.15.5 Wiring. All interior conductors shall be Type TW, fungusproofed. The lead-in cable from the main supply power receptacle on the van exterior to the supply side of the main circuit breaker shall be installed in rigid-metal conduit. Branch circuits for the auxiliary motors and other electrical equipment shall be installed in electrical metallic tubing, except when cable conforming to MIL-C-3432 is specified. For protection against shocks, the grounding circuits shall be as specified herein. All wiring shall be of the surface mounted type within the unit interior, each line shall be restricted to being parallel along walls and ceiling, or at 90 degrees near the corners, to the unit body longitudinal centerline. Circuits of 120 volts, single-phase, shall be provided for lighting and for equipment receptacles. Electrical wire for the circuits shall conform to MIL-W-5086, type as applicable. A clearly marked area, with securing facilities, shall be provided inside the unit for stowing the power supply cables (see 3.15.1 and 3.15.2), when they are not in use.

3.15.6 Grounding. All motor frames and electrical equipment enclosures shall be grounded to the unit frame with bare, minimum No. 6 AWG copper conductor per DOL OSHA regulations and the NEC. The conversion unit electrical system neutrals shall be grounded to each panelboard neutral bus. A bolt with wingnut shall be mounted on the semi-trailer frame in a location where the ground-rod cable can be attached by a man standing on the ground. A ground-rod shall be provided conforming to Figure 4. The ground-rod assembly shall consist of a driving bolt and three, 3-foot lengths of 5/8-inch diameter rod, each with a coupling. Rods shall be made of a high-strength steel core with a copper surface bonded to the core by a fusion. each end of each rod shall be threaded, except that one rod shall be provided with a conical point. Couplings shall be made of copper alloy. The driving bolt shall have a square end and shall be made of steel. All threads shall be 5/8-inch-11 NC. The grounding cable shall be in accordance with NEC and OSHA regulations. Means shall be provided for storing the ground-rod assembly and cable when they are not in use.

3.15.7 Electric motors. Integral horsepower electric motors utilized in the unit construction shall conform to NEMA-STD-MG1, integral horsepower, polyphase, squirrel cage, alternating current (ac), totally enclosed fan cooled continuous duty, speed 1800 rpm nominal electric motor size shall be such that the driven load will not exceed the continuous duty rating of the motor when the unit is operated at the specified production capacity. Electric motor ratings shall not exceed 460 volts 60 Hertz. When V-belt drives are furnished, the belts shall be in accordance with the Rubber Manufacturers Association (RMA) Engineering Standards. Belts shall have a service factor of 1.5. Where reversal of rotation is possible and it could cause incorrect operation of, or damage to, any component of the conversion unit, the motor shall be equipped with a phase-sequence control to prevent operation in the wrong direction.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:06Z
Check the source to verify that this is the current version before use.

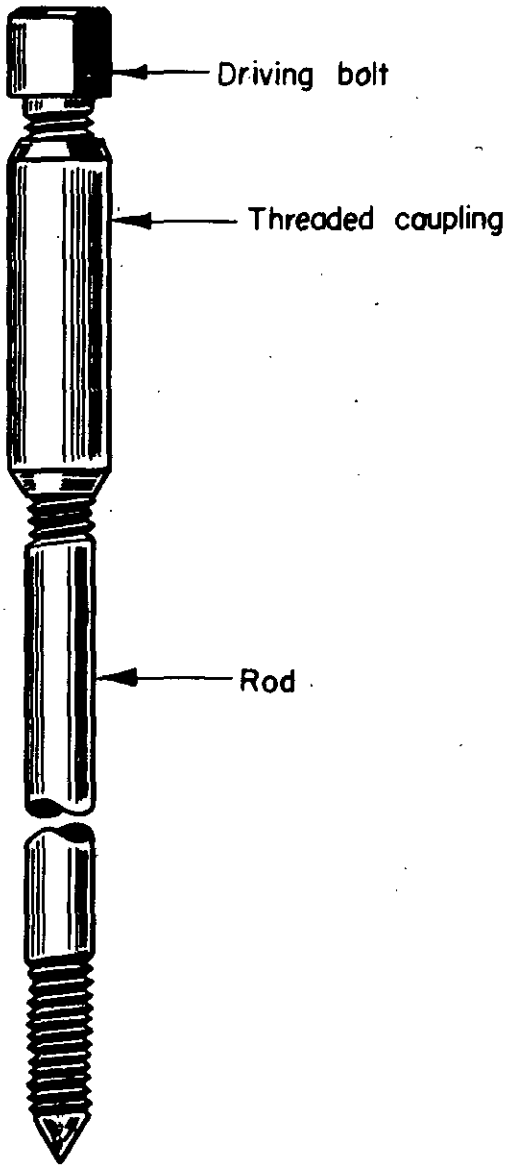


FIGURE 4. **Rod assembly, ground.**

A4463-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

3.15.8 Motor starters. Motors shall be provided with motor starters in accordance with MIL-S-12514, Type I, Class 2 with NEMA Type I enclosure and of voltage and horsepower for the motor specified. Class 1 starters may be utilized for use with the ice saw and water pump. Each motor starter shall be provided with overload and low-voltage protection. The overload protection shall be of the hand-reset type. Fuses shall not be acceptable as overload devices. Compressors, pump, and ice press starters shall have signal lights which will indicate whether or not the starter contactors are closed and energized. Pushbutton stop-start buttons for Class 2 starters for the carbon-dioxide compressor and press shall be mounted on the starter frame and shall project through the box cover. The starter for the refrigeration compressor shall be provided with a 3-way selector switch for hand-off-automatic operation. Panel lights shall be provided to indicate when the motors are energized.

3.15.9 Receptacles. Receptacles, switches, and outlets shall be marked to indicate voltage and their intended use. Two single receptacles for the space heaters and not less than 6 double duplex receptacles 120 volt rating of the UL approved 3-wire type shall be installed. One exterior 120 volt outlet shall be furnished and installed on the outside of the unit body. The outlet shall be of the 3-pole, grounded type with captive chain, waterproof, screw cover. The location of this receptacle shall be in the vicinity of the weather shelter (see 3.16.6). The outlet shall be accessible with the personnel door swung open against the unit body.

3.15.10 Overhead lights. The unit shall be provided with an explosion-proof lighting system that will provide the following illuminations:

(a) For reading gages and operating controls-commercial level-A-100 (70-150 foot-candles).
(b) For the remainder of the unit-commercial level C-20 (15-30 foot-candles).

3.15.10.1 Switches. On-off switches shall be provided for the control of lights.

3.15.10.2 Lamps. Rough service type lamps shall be furnished.

3.15.10.3 Extension trouble lamps. Two heavy-duty extension trouble lamps equipped with bulb guards and cords not less than 25 feet in length shall be provided.

3.15.10.4 Floodlamps. Two clamp-on-type portable floodlamps shall be furnished for providing illumination of the outside and sheltered work area during night operation. Each floodlamp shall operate off the 110-volt electrical circuit and shall be equipped with not less than a 150-watt bulb.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:06Z
Check the source to verify that this is the currentversion before use.

The attachment clamps shall be of the quick-attachment type and shall be securely attached to the semitrailer body door frames, doors, or other convenient supports. Each clamp shall include a ball-and-socket joint which will permit the floodlamp beam to be directed where desired. Each floodlamp shall be provided with a heavy-duty, flexible extension cord of not less than 25 feet in length of sufficient size to service the bulb specified. One end of the cord shall be permanently attached to the lamp. The other end shall be equipped with a disconnect plug for attachment to the outlet receptacles specified. On-off switches shall be furnished to control all lights. The switches shall be mounted in locations accessible to operating personnel. Each floodlamp shall be provided with a manually operated on-off switch either on the lamp body or in the cord adjacent to the floodlamp.

3.15.11 Unit body telephone jack system. A separate telephone jack electrical system shall be provided to interconnect the telephones within the van bodies to the external telephone lines outside of the van bodies. Two telephone jack boxes shall be furnished and shall include standardized commercial binding posts, three posts in each box. The boxes shall be of the nonmetallic type with a minimum size of 1-1/2- by 2- by 4-inches. The binding posts shall be interconnected with coded, electrical wire encased in loom. The boxes shall be equipped with covers, a facia plate on the inside box, and a spring loaded top or forward hinged waterproof cover on the outside box. The telephone line and powerline layout shall preclude static interference on the telephone lines.

3.16 Type I - Semitrailer mounted. For Type I plants, the operating components shall be mounted inside a 30-foot van-type semitrailer as specified herein.

3.16.1 Dimensions and weights. The plant shall conform to the dimension and weight limitations specified in Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

Table I. Semitrailer Mounted Weights and Dimensions

(a)	GVWR, min. (DOT 49CFR567 and TTMA RP No. 39)	Min. 40,000 lb.
(b)	Front coupler weight limit	Max. 48 percent of actual GVW.
(c)	Rear running gear weight limit	Max. 68 percent of actual GVW.
(d)	Overall length	Max. 360 in.
(e)	Overall height	Max. 134 in.
(f)	Overall width	Max. 96 in.
(g)	Ground clearance (SAE J894)	Min. 9 in.
(h)	Ground clearance (exclusive of rear suspension) (SAE J894)	Min. 14 in.
(i)	Landing wheel clearance, radius ("LWC" of SAE J701)	Min. 86 in.
(j)	Swing radius ("SR" of SAE J701)	Max. 58 in.
(k)	Kingpin setting ("KP" of SAE J701)	Range 18-36 in.
(l)	Upper fifth wheel height	52 $\pm$ 1 in.
(m)	Angle of departure (SAE J894)	Min. 25 deg.

3.16.2 Semitrailer specifications. The semitrailer specifications shall be in accordance with Table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

Table II. Semitrailer Specifications

Feature	Requirements
(a) Semitrailer chassis	Provide strength, flexibility, and rigidity to withstand dynamic, over-the-road and off-road transport as well as equipment and machinery operation at operating sites.
(b) Frame	All welded construction; AISI high strength low alloy (HSLA) steel, SAE J410 or ASTM A242, having min. 50,000 psi guaranteed yield point (YP); Stress max. 50 percent of YP; Continuous longitudinal frame members.
(1) Intervehicular connections	Front airbrake and electrical connections per RCCC RP-107; Rear connections per SAE J849.
(2) Upper coupler assembly	Per DOT FMCSR 393.70, TTMA RP No. 6 and RP No. 19, TOFC rated; Kingpin per SAE J700 and J133, 2-in. dia., alloy steel type heat treated in the throat area; Front coupling ramp split pipe or beveled plate and gusset type, min. 2-in. vertical height; Coupler plate AISI HSLA steel, same as frame; Size to cover a 36-in. dia. 5th wheel, min. 88 in. width, and min. 1/4-in. thick; Kingpin reinforcement of eggcrate type; External SAE J534 lube fittings with lines to lubricate both road and curb sides of plate when coupled to truck-tractor.
(3) Front supports (landing gear)	Per TTMA RP No. 4 including TOFC operation; Twin vertical lift, telescopic, non-rotating type with two speed gearbox and folding handcrank with holder; Leg travel, min. 17-in., providing upper coupler-to-ground clearance at kingpin of at least 46-in. through 58-in.;

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:06Z
Check the source to verify that this is the current version before use.

Table II. Semitrailer Specifications (Cont'd)

Feature	Requirements
	Full skirt type support legs; Fore and aft longitudinal bracing and transverse bracing; Legs equipped with equalizing mud shoes having min. 140 sq. in. area for each shoe; Edges turned up; Supports withstand TOFC, Marine transport, and free-fall uncoupling drop of at least 4-in. height without permanent deformation; Gearboxes filled with reopectate silicone base grease, intended for cold weather operations per RCCC Advisory AV 7-5.
(c) Suspension system	Tandem air spring (2-axle) type intended for on-off road severe use; Withstand both dynamic and stationary loads and vibration; Min. 8-in. travel range each axle from compression to rebound; Cushioned bumper stops, anti-sway components linking roadside and curbside and ride dampening heavy-duty shock absorbers; Air spring suspensions of trailing member type installed with axles in correct alinement with kingpin and semitrailer centerline per both suspension and axle manufacturer's recommendations; Piston or multiple bellows per SAE J551, having internal bumpers; Air pressure safety regulator valve, air controls, and air-tight plumbing; Ride frequency max. 130-cpm at center of interior floor; Height control valves, closed center time delay type, protected from road splash, debris, and obstacle damage; Time delay limits min. 3-sec. to max. 8-sec.;

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:06Z
Check the source to verify that this is the current version before use.

Table II. Semitrailer Specifications (Cont'd)

Feature	Requirements
	Guarded driver over-ride air controls on roadside front for selection of (1) maximum suspension height, (2) normal automatic height, and (3) low, depleted-air height; Plumbing of flexible hose and reusable swivel fittings same as for air-brake system.
(1) Axles	Per DOT FMVSS and TTMA RP No. 39, 2-ea. having GAWR min. 20,000 lb. hub rating; Min. 20,000-lb. brake rating and min. 25,000-lb. bearing and axle beam rating for each axle; Hardened and ground alloy steel spindles; Standardized tapered roller bearings with cups, bearings, and cones in matched sets, all from the same bearing manufacturer for each vehicle; Lube seals per TTMA RP No. 43 w/visual cap windows; Axle hubs per SAE J694, 10 stud, 11-1/4-in. BC, with out-board mounted brake drums when brake drums are furnished.
(2) Tires and wheels	All components per TRA Yearbook and DOT FMVSS; 5-ea., single, tire and heavy duty disc wheel assemblies; Rim and wheel disc fabricated by same manufacturer or as recommended and approved by rim manufacturer; Wheels with a standard disc offset (dish).
(3) Tires	Widebase single tubeless type; Size min. 18-22.5, min. LRH (16 PR) having road lug tread design; Cold inflation matched to loading per TRA Yearbook, max. 80 psi; Metal valve caps w/elastomer seals.
(4) Balancing	Tires, wheel, hub, and drum or disc, all inherently in balance and not exceed max. 0.050-in. radial and lateral run-out (TIR) or max. 120 in.-oz. out-of-balance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

Table II. Semitrailer Specifications (Cont'd)

Feature	Requirements
(5) Spare	An inflated spare tire and wheel assembly furnished with each vehicle.
(d) Rear components	Van wheel housings and container semi-trailer fenders and splash guards per SAE J682, SAE J683, and TTMA RP No. 41; Fenders of lighweight aluminum or fiber-glass reinforced plastic designed to support a min. 400-lb. load on any 1-sq. ft. area at any one time; Splash guards anti-sail type of cold-weather type elastomer having a min. 3000-psi ultimate strength; Guards mounted by metal clamping straps fastened with hex bolts and nuts through RCCC/TTMA standard holes; Rear tow hook per SAE J849 reinforced to frame; Rear bumper per DOT FMCSR and TTMA RP No. 10; Horizontal dock bumper, min. 8-in. height, having min. 4 each elastomer blocks on bumper face; Retractable, 2-position hinged lower DOT bumper mandatory, if the trailer's SAE J894 angle of departure at centered, normal travel loaded height level exceeds a 28 degree angle; Fixed bumper or retracted position of hinged bumper adequate to provide clearance for travel without ground contact under off-road conditions.
(e) Brake system	Per DOT FMVSS, DOT FMCSR, and California Vehicle Code; Straight air service and emergency-parking system; Withstand high speed, off-road, and mountainous travel and creek bed immersion usage; Provide priority to brake system over any auxiliary air system use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

Table II. Semitrailer Specifications (Cont'd)

Feature	Requirements
(1) Gladhands	Per SAE J318 color coded yellow for control (service) on curbside and red for supply (emergency) on roadside; ID tags and spring-loaded gladhand dustcovers furnished; Locations 2-each front per RCCC RP-703 and 2-each rear per SAE J849.
(2) Relay valves	High speed type; Waterproof components.
(3) Air reservoirs	Min. 3-each per DOT and SAE J10; 1 each service, 1-each parking release. and 1-each air suspension; Sized per DOT and component manufacturer's recommendations; Automatic drain valves on each reservoir; No component lower than bottom of frame rails.
(4) 121 Anti-lock system	When legally required, per DOT and RCCC RP's; Hermatically sealed and water-immersion-proof electrical components; Min. 80 percent of anti-lock systems electrical cable length covered or encased by grounded metal (including frame members) or system filtered to prevent external radio and electro-magnetic interference; TTMA RP No. 42 indicators located near axles; Indicators, running lamp with bulbs and red lenses per RCCC RP-108 as furnished on semitrailer; Twin bulb running lamps may have one bulb removed to cut down on the amperage draw for the computer; Axle tube holes sealed waterproof by grommets and RTV silicone sealant around sensor wires.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:06Z
Check the source to verify that this is the current version before use.

Table II. Semitrailer Specifications (Cont'd)

Feature	Requirements
(5) Foundation brakes	Disc type or wedge or S-cam type with removable dust shields; Means to prevent simultaneous service brake and emergency parking brake application from causing any damage to the foundation brakes, an interlock is acceptable; Components protected from damage, located above the bottom of the brake disc or brake drum; Disc, wedges, cams, and roller pins, as a minimum, stainless steel or hard chrome plated and silicone high temp compound on bearing surfaces to prevent corrosion malfunction; Drum type brakes equipped with FMS Automotive Data Book lining bolted to the shoes and with outboard mounted drums.
(6) Brake actuators	Heavy-duty spring type having environmental protective covers, boots, and seals; Elastomer coated main springs capable of holding wheels locked on grades up to min. 30 percent; Self-adjustors for each wheel, providing correct compensation for lining wear without lock-up or unbalance; Manual means to adjust in case of malfunction.
(7) Plumbing	All hose type per RCCC RP-305 having stainless steel reinforcement and reusable swivel fittings permitting hose replacement from bulk hose supply; Routing within protection of framing members; Elastomer cushioned clamps and grommets or thru-frame connectors to secure and support hose; Identify hose function adjacent to each fitting with waterproof means; Min. 3-in. of slack at relay valves.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

Table II. Semitrailer Specifications (Cont'd)

Feature		Requirements
(f)	Vehicle electrical System	Per DOT, CA Code, SAE Handbook, TTMA RP No. 9, TTMA RP No. 14, and RCCC RP Manual; Heavy-duty waterproof system providing protection from physical damage, shock, vibration, and environment; System 12-volt potential with negative ground per SAE J538.
	(1) Front receptacle	Per RCCC RP-703 and similar to TTMA RP No. 6; Receptacle per TTMA RP No. 40 and RCCC RP-107 with elastomer truck-tractor plug support bracket; Weatherproof connection box with nut and stud terminals; SAE J553 circuit breakers protecting all live circuits, no fuses; If used, receptacle-to-remote-box cable per SAE J1067.
	(2) Rear receptacle	Same as front receptacle and to include dolly converter plug support bracket; Location per SAE J849.
	(3) Wiring	Per SAE J163, SAE J555, and RCCC RP Manual; Wire identification per SAE J560 or, optionally, SAE J821; Multiple circuits combined into harness; Wiring supported at max. 36-in. spacing with elastomer ties and secured mounts to prevent whipping, chafing, and mechanical or fatigue damage; Interior of splicing devices waterproofed with RTV silicone sealant; RCCC RP-110 wire with RCCC RP-112 terminals; Main junction blocks of nut and stud type having an electrical varnish coating after assembly; Wire-to-wire terminals secured to vehicle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

Table II. Semitrailer Specifications (Cont'd)

	Feature	Requirements
(4)	Marker lamps	Per RCCC RP-108/1, RP-108/2, or RP-108/3, "preferred" types, waterproof and shock resistant; Locations per RCCC RP-702 and TTMA RP No. 9, including reflectors; Reflectors secured to vehicle with metal fastener(s).
(5)	Rear combination lamps	Per SAE J759, SAE-IST, and TTMA RP No. 14, flush-mounted elastomer housing w/bulb and snap-on or elastomer gasket lens or sealed unit w/elastomer ring; Round 4-in. dia. size installed in TTMA standard 4-1/2-in. hole; Double filament configuration with 3-wire pigtail, including ground to frame; Weatherproof and shock resistant; Total 4-each; 2-each outboard Turn and Clearance and 2-each inboard Stop and Tail Functions.
(6)	Lamp bulbs and lenses	Hermatically sealed units or fleet service heavy-duty bulbs; Bulbs mounted in vibration-protected sockets w/ceramic or plastic insulators and stainless steel or elastomer bulb tension devices; Lenses of polycarbonate plastic.
(7)	Wiring diagram	Metal plate, showing wiring furnished, per TTMA Technical Bulletin No. 65 and "Metal Name Plate Industry Standards and Practices" per NAME.

3.16.3 Military vehicle electrical system. When specified (see 6.2), the plant shall be equipped with a military compatible 24-volt service lighting system to provide towing capability for the military design trucks as well as commercial trucks. In addition to the TTMA 7-contact receptacle, the plant shall be equipped with a military 24-volt, 12 contact intervehicle receptacle in accordance with MS75021-1, having a spring-loaded, waterproof cover. The military receptacle shall be located near the TTMA receptacle and both shall be labeled with identication plates for 24-volt and 12 volts respectively, negative ground. The military 24-volt wiring shall be routed

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

into a weatherproof stud and nut type junction box, equipped with SAE J553 circuit breakers, protecting live circuits B, E, and J of the military service lamp portion. All unused leads of the cable shall be insulated and sealed watertight. The military 24-volt circuits shall be connected through replaceable resistors, for live circuits, and to the trailer frame, for ground circuits, in accordance with Table III.

Table III. Military-Commercial Connections

Function	Mil. Receptacle Terminal	Connection Instructions	Comm. TTMA Wiring Ident.
Blackout Marker	A	No.	-
Ser. Left Turn & Stop	B	Connect to -	YEL (3)
Blackout Marker	C	No.	-
Ground	D	Connect to -	WHT (1)
Ser. Tail & Marker	E	Connect to -	BLK (2)
		& Connect to -	BRN (6)
Blackout Stop	F	No.	-
Blackout Marker	H	No.	-
Ser. Right Turn & Stop	J	Connect to -	GRN (5)
Aux. Power (Hot)	K	No.	BLU (7)
Ground	L	Ground to Frame	-
Electric Brake	M	No.	-
Electric Brake	N	No.	-

With the military system installed, the TTMA marker circuits, color coded BLK (2) and BRN (6), shall be electrically isolated from each other, thus preventing 12-volt current cross-over when operated by commercial trucks through the 12-volt TTMA plug. Replaceable diodes may be used, having a minimum amperage of 150 percent of the current demand of the running lamps served. Military circuits letter coded B and J on the tactical trucks are passenger car combination stop and turn signal type, using the same lamp filament, and there is no single stop lamp circuit. As a result, the plant's turn signal lamp circuits, YEL (3) and GRN (5), shall function as both turn signal lamps and stop lamps and the plant's normal stop circuit, RED (4), cannot be energized from the military 24-volt system input.

3.16.4 Semitrailer van body specifications. The semitrailer body specifications shall be in accordance with Table IV.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:06Z
Check the source to verify that this is the current version before use.

Table IV. Semitrailer Van Body Specifications

Feature	Requirements
(a) Reefer ratings	Per TTMA RP No. 38 for manufacturer's commercial model of van body supplied; Heat transmission rate, max. 25 Btu/h ° F, for each 10-foot increment of van body OAL (overall length); Van body air leakage rate per hour in CFH max. 65 percent of interior van body volume prior to installation of equipment.
(b) Framing	AISI HSLA steel members; Aluminum alloy extrusion members; Plastic galvanic action separators furnished; Framing members and construction to prevent a direct metal-to-metal thermal bridge between inside and outside of van body.
(c) Exterior panels	Smooth skin type, waterproof, air-tight, and stress-wrinkle free; Panel material either fiberglass reinforced plastic (FRP) with plywood core or aluminum sheeting intended for containers and semitrailers; Provide FRP plywood per APA Plywood Design Specification Manual X215 and U.S. Product Standard PS 1; Core of APA Group 1, Exterior Type; Panels min. 1/2-in. thick on front, sides, and roof panel having laminate of min. 22-oz./sq.yd. material or woven roving (W.R.) or combination of both; Ultra-violet-inhibitor gel coat on exterior of panel having min. 0.015-in. thickness; Exposed edges of plywood treated or dipped to repel moisture; Aluminum sheet, min. 0.05-in. thickness; Inside surface of skin correctly degreased and coated with primer per manufacturer to insure foam insulation adhesion;

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:06Z
Check the source to verify that this is the current version before use.

Table IV. Semitrailer Van Body Specifications (Cont'd)

Feature	Requirements
	Seamless, 1-piece roof sheet with adhesive bow fastening and perimeter attachment, sealed to prevent moisture from entering van body; Exterior roof load, min. 300 lb/sq.ft. (for semitrailer) or min. 660 lb/sq.ft. (for ISO container) imposed over any 2-sq. ft. area at any one time and min. 40 lb./sq. ft. snow load over the whole roof.
(d) Insulation	Closed cell, rigid foam, such as isocyanurate, polybenzimidizole or polyurethane foam; Fill and bond spaces between exterior panels and interior panels, without voids; Nominal thicknesses, min. 4-in. roof, min. 3-in. sides and front, and min. 2.7-in. under floor; Density nominal 1.8-lb/cu. ft.; Rated "Nonburning" per ASTM D 1692 or D 2863 having an equivalent oxygen index; Foam to adhere to van body skin with an adhesive force greater than the co-adhesive strength of the foam; After foam-in-place, max. 5/8-in. of bow and warp in exterior and interior panels when measured with min. 4-ft. length metal straightedge.
(e) Interior panels	Smooth, flat refrigerator type FRP panels; Panels having laminate of min. 18 oz./sq. yd. material or woven roven (W.R.) or a combination of both; Front and sides min. 5/16-in. thick and min. 1/8-in. thick FRP sheet on ceiling; FRP panels of low fire hazard type having min. UL Class B Fire Hazard Classification; Max 50 flame spread, max 5 fuel development, and max 75 smoke development of panels per ASTM E-84;

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

Table IV. Semitrailer Van Body Specifications (Cont'd)

Feature	Requirements
	All panel joints and edges sealed, using silicone sealant to prevent entry of moisture into insulation; Panels fastened with largehead plastic drive rivets (and for ISO containers, TIR fasteners at corners); Wall post and front post location indicators furnished on ceiling and floor to provide equipment installer with actual mount reference for installation of operating fasteners.
(f) Floor	Per TTMA RP No. 39 for forklift trucks having min. 12,000 lb. axle load; Cross members evenly spaced along length of van body at max. 12-in. on centers; When interior floor of metal, aluminum nonskid tread type; When interior floor covered, floor covering per L-F-475, Type II, Grade A, installed on subflooring with waterproof adhesive per floor covering manufacturer; Color of floor covering to compliment wall color.
(g) Windows	Double glazed fixed type; Glazing per DOT FMVSS and ANSI Z26.1; Edged with flexible elastomer channels; Provisions for pane replacement; Desiccant material and high altitude breather between panes; Removable exterior expanded metal guards; Interior molding of light weight metal or elastomer around inner perimeter of each window between window and interior panels.
(h) Doors	Provide double, van body doors at both front and rear ends and min. 2 each personnel doors in the sides; Door hardware, noncorrosive marine container type;

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

Table IV. Semitrailer Van Body Specifications (Cont'd)

Feature	Requirements
	Air sealing flexible gaskets, not subject to damage by personnel foot traffic; Hold-open catches and door bumpers; Anti-drip rain protection over each door.
(1) Side doors	Size min. 30-in. wide and min. 69-in. high; Forward exterior edge hinged with full height piano hinge having min. 1/4-in. stainless steel hinge pin; Latching mechanism featuring top and bottom securing with roller and ramp operation; Inside and recessed outside door handles; Lockable from outside with a padlock having a min. 3/8-in. shackle.
(2) Double doors	4 each total; 2 each on front end and 2 each on rear end; Recessed for protection by corner posts; With both doors open, clear opening min. 86-in. wide and min. 66-in. high front and min. 76-in. high rear (for semitrailer) or min. 76-in. high front and rear (for ISO container); Door panels of either FRP plywood or plymetal foam sandwich construction; FRP plywood per US PS1 and APA Group 1, min. 3/4-in. thick with min. 24-oz./sq.yd. fiberglass composite; Edges of plywood treated or dipped to repel moisture; Ply metal, min. 1-in. thick aluminum faced plywood composite laminate having min. 0.03-in. bonded exterior and interior aluminum sheeting; Hinge points solid core to prevent hinges from being fastened through any imperfections;

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:06Z
Check the source to verify that this is the currentversion before use.

Table IV. Semitrailer Van Body Specifications (Cont'd)

Feature	Requirements
	Doors hung on corner posts by min. 4 each heavy intermodal container door hinges with stainless steel pins; Doors secured with cam type upper and lower container latching mechanisms, 2 each per door panel.
(3) Dry-ice removal side door	Door min. size 15-in. x 15-in.; Location in curbside van body wall immediately back of bandsaw and scale area; Door hinged at outside top edge w/full length piano hinge having min. 1/4-in. dia. stainless steel hinge pin; Provide twin, door hold-open braces of the automatic hold and release type; On inside only, provide a twist handle to operate 2-ea. slam latches, w/strikers, that are positioned in the lower door corners; For the door, provide a dry-ice chute complete w/stanchons and anchor brackets, leading from the dry-ice removal side door to a MIL-T-40019 storage tank on the ground inside of the weather shelter.
(i) Tailgate	Rear, bottom hinged furniture van type for semitrailer; Min. 86-in. wide and min. 50 percent of rear door height; Fold vertically against doors and lock for travel and open horizontally and lock for operation; Top covered with nonskid aluminum tread sheet or plate.
(j) Boarding steps	Per DOL OSHA regulations and SAE J185; Light metal boarding steps with at least 1 each removable handrail installed away from door hinges and not interfering with opening swing of doors; Ladder mounts of semi-permanent type, preventing instability under heavy personnel traffic; Steps and handrails secured inside for travel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

3.16.5 Weather shelter. The unit shall be equipped with a weather shelter as specified herein. The shelter materials and fabrication shall be comparable with the materials and the fabrication specified in MIL-C-13489. Provisions shall be made for onboard storage and transport of the shelter and hardware. A fabric canopy and shelter extending from the curb-side rear to the front side of the entrance door shall be provided to protect the one-ton ice container and operating personnel from the sun, storms, and inclement weather. Provisions shall be made for securing the canopy roof and walls to the unit body. The canopy shall be of such width as to provide 3 feet of work space around the ice chest and the height from the ground shall be not less than 9 feet at the lowest point. Side walls of the same material as the canopy shall be furnished. Tie back-openings shall be provided in the side wall of the shelter for freedom in moving the ice in and out. Metal tent stakes of sufficient number to secure the side walls to the ground at not more than 4-foot intervals shall be provided. All necessary materials and hardware shall be provided for secure installation of the canopy and side walls in place. A fabric wall shall be provided to cover the space between the unit body and the ground. A collapsible duct shall be provided for evacuating the exhaust from the ventilating fan outside of the shelter area. A warning shall be provided to require use of the duct when the shelter is in use. Fasteners shall be provided for attaching the duct to the ventilator outlet.

3.17 Type II, ISO container mounted. For Type II conversion units, the operating components shall be mounted inside the ISO container van bodies. The van bodies shall be modified commercial, refrigerated type freight containers constructed identical to the container manufacturer's, Series 1, ISO freight container product line, which comply with the following requirement documents:

(a) ISO Recommendation No. 668.
(b) ISO Recommendation No. 1496.
(c) ANSI MH5.
(d) Transport International Routiers (TIR) Cornets (49CFR420).
(e) International Safe Containers (49CFR450).
(f) American Bureau of Shipping (ABS) Guide for Certification of Cargo Containers.

Unless otherwise specified, the conversion units shall be mounted in 2 each, size 1C, 6 000 mm (20 ft) ISO containers. When specified (see 6.2), the conversion unit shall be mounted all in 1 each, size 1A, 12 000 mm (40 ft) ISO container. The manufacturer of the ISO container van bodies shall have fielded to commercial users, TIR and ABS certified, Series 1, Size 1A and 1C, ISO containers for commercial intermodal land and sea service prior to procurement solicitations for these conversion units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

3.17.1 Container specifications. The ISO container used in the Type II unit shall comply with the Type I requirements for the semitrailer van body specified in 3.16.4, the additional container specifications herein, and Table V. The container framing shall be in accordance with ISO and ANSI requirements. The corner fittings shall comply with ISO minimum lifting forces. Provisions for straddle carrier container handling shall be furnished. Boarding steps shall be sized for the unit when mounted on the container semitrailer. The semitrailer tailgate shall not be furnished on the rear of the container.

Table V. Container Mounted Weight and Dimensions

Characteristic	Requirement
(a) Weight, container with operating equipment installed, maximum each	25,000 lb. (size 20) 45,000 lb. (size 40)
(b) Container, exterior dimensions (for ISO intermodal transport):	
(1) Length	ISO/ANSI (Ref. 240.0 in. or 480.0 in.)
(2) Height	ISO/ANSI (Ref. 96.0 in.)
(3) Width	ISO/ANSI (Ref. 96.0 in.) 1/
(c) Container interior:	
(1) Length, minimum	200 in. (size 20) 440 in. (Size 40)
(2) Height, minimum	80 in.
(3) Width, minimum	84 in. 1/

1/ When specified (see 6.2), the overall exterior container body width shall conform to the proposed or approved ISO width limit of 102 inches (2.59 meters) and the inside width shall provide a corresponding increase.

3.17.1.1 Compatibility. The container shall be compatible with the commercial ISO freight container semitrailers for the 30-foot-long, 1C container size and shall be compatible with applicable military semitrailers, including semitrailer, lowbed, 40-ton, 3-axle, M870, commercial construction equipment. The unit shall be acceptable to the domestic and international transport carriers for unrestricted movement of ISO freight containers in the interchange between rail, marine, and highway carriers models and worldwide trailer operation shall be acceptable to the domestic and international transport carriers for unrestricted movement of ISO freight containers in interchange between rail, marine, and highway carrier modes worldwide container operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

3.17.2 Lifting and tiedown provisions. Each size 40 unit container body shall be equipped with four removable lifting and tiedown mechanisms for handling of the van bodies by cranes when demounted off the vehicle. The van body end of the mechanism shall be compatible with the van body ISO container corner fittings and shall be of the ISO safety bayonet type. The mechanism bayonet engagement in the locked position shall be of the manual twist lock type having spring-loaded detent positions or a positive slip lock sleeve. The locked position shall be either apparent to personnel or identified by marking. The crane hook end of each mechanism shall be provided with an eye or clevis having an inside clearance diameter of not less than 3 inches. Each removable lifting and tiedown mechanism shall be rated by the mechanism manufacturer to withstand the loads imposed for use with the largest and heaviest commercial ISO freight container. When not in use, the mechanisms shall be stowed in the tool compartment.

3.17.3 Container semitrailer. When specified and as specified (see 6.2), a container semitrailer for each unit ISO container van body shall be furnished. The semitrailer shall be of the skeleton type with heavy box beam bolsters having ISO bayonet twist locks. The center of each twist lock shall be dimensionally identical to ISO No. 668 for applicable size of containers. The front twist lock shall engage either in the bottom or in the front of the container and shall provide the load restraint in compliance with DOT FMCSR and TTMA RP No. 47. The container semitrailer shall be equipped with the semitrailer component specifications specified in 3.16, Tables I, II, and, when specified, III, with the additions and exceptions specified herein. The semitrailer shall be equipped to transport any ISO Series 1 container, including the unit ISO container van bodies. The container semitrailer shall be in accordance with Table VI. The semitrailer shall be equipped with a tridem (3-axle) air suspension system providing not less than 60,000 pound rating. Three each minimum 20,000 GAWR axles, and 7 each wide base tire and wheel assemblies. The semitrailer shall be furnished with a rear nonskid platform in substitute of the folding rear tailgate required on the Type I semitrailer van body (see 3.16.4(i)).

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:06Z
Check the source to verify that this is the currentversion before use.

Table VI. Container Semitrailer Weights and Dimensions

		For size 1C (20 footers)	For size 1A (40 footers)
(a)	Semitrailer payload rating	Min. 20 tonnes (Ref. 44,800 lb.)	Min 30 tonnes (Ref. 67,200 lb.)
(b)	GVWR (TTMA RP No. 39)	Min. 20 tonnes plus semitrailer curb weight	Min. 30 tonnes plus semitrailer curb weight
(c)	Front coupler weight	Max. 48% of GVWR	Max. 48% of GVWR
(d)	Rear running gear	Max. 68% of GVWR	Max. 68% of GVWR
(e)	Overall length (OAL)	Max. 346-in.	Max. 540-in.
(f)	Overall width	Max. same as container in Table V, (b), (3)	Max. same as container in Table V, (b), (3)

Note. A tonne = 1 000 kg = 2240 lbs.

3.18 Electromagnetic interference. The electromagnetic interference (EMI) emission characteristics of the unit shall conform to MIL-STD-461, for electrical/electromechanical equipment with intermittent short duration interference acceptable.

3.19 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.19.1 Lubricants. The conversion unit shall operate as specified herein when lubricated with military lubricants (see 6.4).

3.19.2 Lubrication fittings. Lubrication fittings shall conform to SAE J534 and shall be of the threaded types. Fittings shall be located in a protected position and shall be accessible to a minimum 15 ounce, hand operated grease gun with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility. A pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts. A lube chart shall be furnished on each unit in accordance with SAE J753. Oil holes shall be equipped with spring-loaded covers.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:06Z
Check the source to verify that this is the current version before use.

3.19.3 Filling, draining, and checking provisions. Enclosures such as gear cases and transmission housing that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with a lubricant and for draining. The drain outlet shall be fitted with a drain plug conforming to MS35844 or MS49006. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the conversion unit is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to the ground.

3.19.4 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

(a) MIL-G-23827 - For instruments, all sealed bearings, and other bearings where operating temperatures do not exceed 250° F.
(b) MIL-G-10924 - For bearings where operating temperatures do not exceed 175° F.

The unit shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not compatible with other greases. Each component of the complete unit shall be itemized on a tag in a conspicuous place to indicate which military grease has been used.

3.19.5 Oil lubrication. Unless otherwise preserved for packaging (see 5.2.1), all components shall be filled to the operating level with winter grades of military oils designated for use in the temperature range of 0° F to 32° F. Each component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.19.6 Servicing and adjusting. The contractor shall service and adjust each unit for operational use including at least the following:

(a) Conversion unit:
- Adjustment and alinement of mechanical components.
- Adjustment of heating system.
- Electrical system continuity checks.
- Lubrication of components.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

(b) Semitrailer, when specified or furnished:
- Adjustment of suspension.
- Adjustment of brake system.
- Electrical system continuity checks; voltage drop checks.
- Torque of wheel lug nuts.
- Inflation of tires.
- Lubrication of components.

3.20 Fungus and moisture resistance. The electrical circuitry of all electrical systems, including all components and connections except as specified below, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.20.1, with 1 percent copper-8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.20.1 Composition. Composition I shall be used unless Composition II must be used in order to comply with local air pollution regulations in the application of the varnish. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 102 - Southern California Air Pollution Control District.

3.21 Fire extinguishers. The unit shall have space provisions for installation of two, 15-pound, carbon dioxide fire extinguishers with fire extinguisl brackets. Space shall be provided by the contractor so that upon installation of these fire extinguishers, they will be accessible to the operator in the following locations:

(a) One on the road side wall ahead of the unit heaters.
(b) One on the curb side wall, rearward of the CO_2 compressor.

When specified (see 3.32 and 6.7), the fire extinguisher assemblies shall be furnished for contractor installation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

3.22 Lifting attachments. With the exception of the ISO containers, permanently affixed attachments, that enable the other units to be lifted in their normal travel position, shall be provided and shall conform to MIL-STD-209. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.23. Lifting attachments for the ISO containers shall be in accordance with 3.17.2.

3.23 Tiedown attachments. Permanently affixed attachments, that permit fastening the unit to the floor or deck of a transportation medium, shall be provided. The tiedown attachments shall conform to MIL-STD-209. Attachments shall be rounded and beveled to prevent cutting of the tiedown material.

3.24 Railroad humping. The unit shall withstand a railroad humping test of 8 mph in one direction and 8 mph in the opposite direction without permanent deformation to any of its components when tested in accordance with 4.6.2.11.

3.25 Toolbox. The unit shall be provided with a toolbox made of metal not less than 0.125 inch in thickness. The toolbox shall have an overlapping hinged top lid or side door which prevents the entry of rain or dirt, not less than two tension latches, and a hasp for locking with a padlock having a 3/8-inch shackle. The toolbox shall be securely mounted behind the CO_2 press. Access shall be through the double rear doors. The size of the box shall be not less than 36 cubic feet. The lid or doors shall open at least 90 and not more than 120 degrees from the closed position and shall remain in the fully opened position by weight of the lid or door restrained by a chain or other holding device. Provisions shall be made to allow complete draining of the toolbox.

3.26 Identification and marking.

3.26.1 Identification plates. The unit shall be identified in accordance with MIL-STD-130. The marking shall be applied to the unit on plates conforming to MIL-P-514, Type I, Style 1, Composition C, of Type I, Grade A, Class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.26.2 Shipping data plate. A shipping data plate shall be furnished and shall conform to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material. In addition, the plate shall show the silhouette of the unit in transport position indicating the center of gravity and the location and capacity of the lifting and the tiedown attachments. The plate shall be attached by screws, bolts, or rivets in a conspicuous protected location on the curb side of the plant.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:06Z
Check the source to verify that this is the currentversion before use.

3.27 Instruction plates. Each unit shall be equipped with instruction plates or diagrams, including warning and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing of the unit. Instruction plates shall conform to ANSI Z35.1 and shall be in accordance with the "Metal Name Plate Industry Standards and Practices", as published by the National Association of Metal Name Plate Manufacturers, Inc., (NAME). The plates shall be attached in a conspicuous protected location following the securement methods in the NAME Standards and Practices.

3.27.1 Flow diagrams. A flow diagram illustrating each process of the plant, and a flow diagram illustrating the overall process of the unit shall be permanently affixed by screws, bolts, or rivets near the controls in the machinery compartment. The flow diagrams shall be of a size to be clearly readable.

3.27.2 Wiring diagram plates. Electrical wiring schematic diagrams for the plant nonvehicular, electrical circuits shall be affixed near the controls in the machinery compartment by screws, bolts, or rivets. The plates shall be of a size to be clearly readable.

3.27.3 Operating instructions. Operating instructions shall be furnished and shall be permanently affixed by screws, bolts, or rivets near the control panel in the machinery compartment. Instructions shall be clear and concise and shall contain sufficient information to enable operation without damage to the unit or injury to personnel.

3.27.4 Valve marking. Each control valve and safety valve shall be identified by a plate on or near the valve indicating a flow-control number. Plates shall be attached by screws, bolts, or rivets.

3.28 Contact surfaces of dissimilar metals. Contact between dissimilar metals shall be avoided when practicable. The definition of dissimilar metals shall conform to MIL-STD-889. When such contacts could occur, they shall be permanently separated a minimum of 2-mils insulation by the use of joint fillers composed of zinc chromate, nonconductive aluminum powder, inert filler, elastomer gasket, electrical tape, or membrane strips. Asphaltic-type compounds shall not be used nor shall tape and membrane strips of the type that bunches between the metal surfaces during hole drilling operations.

3.29 Treatment and painting. The portions of the unit and its components and parts normally painted in accordance with the contractor's commercial practice shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A. Unless otherwise specified (see 6.2), the exterior finish of the plant shall be Color No. 24087, the finish for the interior walls shall be Color No. 24410, the finish for the unit ceiling shall be Color No. 27875,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

the finish for the unit floor shall be Color No. 26152, and the finish for the machinery shall be Color No. 16187. All colors shall conform to FED. STD. No. 595. Pipelines shall be color coded in accordance with MIL-STD-101. Nonpermissible galvanic couples (dissimilar metals) as defined in MIL-STD-889 shall be insulated as specified in MIL-STD-889.

3.30 Items to be furnished with the conversion unit. No shipment of conversion units shall be made unless repair parts, maintenance tools, operating tools, technical publications, and accessories are included with the shipment or unless approval for shipment of the conversion unit without such items has been received from the contracting officer (see 6.11).

3.31 Marking. Registration number (see 6.5), and other markings of the unit shall be in accordance with MIL-STD-642. The GVWR shall be marked on each forward side of the unit and the correct tire pressure shall be marked on the side, above each tire "TP---", in lettering of a color and style in accordance with MIL-STD-642.

3.32 Government furnished property. When and as specified, the following property in the quantities indicated will be furnished by the Government and mounted by the contractor (see 6.7):

Item No.	Description	Identification	Quantity for each unit
1.	Fire extinguisher CO_2, 15 lb.	NSN 4210-00-202-7858	2 (opt.)
2.	Brackets, fire extinguisher	NSN 4210-00-555-1283	2 (opt.)
3.	Padlocks with two extra keys	NSN 5340-00-682-1504	as required
4.	Publications holder	NSN 7520-00-559-9618	1

3.33 Vehicle weight classification. The contractor shall complete the Government furnished vehicle weight classification forms and apply the vehicle weight classification number(s) to the unit as specified (see 6.6), in the location specified by the contracting officer. The numbers shall be permanently marked, using the size required in MIL-S-40626.

3.34 Workmanship. All workmanship of each van body shall be in accordance with engineering, manufacturing, and production standards of the truck body industry or ISO reefer container industry, as applicable. The van bodies shall be free from workmanship deficiencies that could impair the operation or serviceability of the van bodies. All parts, components, and assemblies of the van bodies including bearings, seals, machinery, stampings, welded parts, castings, forgings, and machined work, shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux and other harmful, extraneous material. External surfaces shall be free of burrs and sharp edges and corners, except when sharp edges and corners are functional.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:06Z
Check the source to verify that this is the current version before use.

3.34.1 Fabrication.

3.34.1.1 Aluminum and steel fabrication. Aluminum and steel used in the fabrication of the van bodies shall provide original quality surface finish and shall be free from kinks and sharp bends. Aluminum and steel having eroded surfaces are not acceptable. The forming of the material shall be done by methods that will not cause damage to the metal. Shearing, punching, and chipping shall be done uniformly, neatly, and accurately. Corners shall be square and true, and all sharp edges and burrs shall be removed. The burned surfaces of flame-cut material shall be free of burrs, slag, and sharp edges. Precautions shall be taken to avoid overheating of metal. Heated metal shall be allowed to cool slowly, except where heat treatment is required. All bends shall be made with precise, unyielding dies or jig fixtures to insure uniformity of size and shape.

3.34.1.2 Fiber glass fabrication. Fiber glass reinforced plastic (FRP) and FRP coated semitrailer plywood used in fabrication of the van bodies shall be smooth surfaced and free from bubbles, glazing cracks, and discontinuities and shall have a uniform FRP thickness. All edges and openings of FRP coated plywood shall be completely sealed to prevent water intrusion between the laminates. The FRP shall be free of parting agents which impair paint adhesion. Welding, bonding, connections, and fabrication of the FRP shall be in accordance with the recommended methods of the manufacturer of the material for this application.

3.34.2 Welders and welding.

3.34.2.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 - Structural Welding Code, Section 5, Qualifications.
Welding Qualification of the ASME, Section IX.

3.34.2.2 Metal welding. All metal welding shall be in accordance with the American Welding Society's Welding Handbook and as specified herein. Welds shall provide the maximum notch toughness and fatigue strength with a minimum of residual stresses in accordance with the Welding Handbook. Except for vertical welding by the electroslag process all welds made on high-stress members shall be made from core or heavy-coated electrodes, gas-shielded, or submerged arc. Electroslag and electrogas processes, when available, shall be included for use on heavy sections, 1 inch or more in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

thickness. Plates and shapes having thicknesses of 3/4 inch or less shall require welds of 100 percent penetration for butt welds and not less than 67 percent penetration for fillet welds. Welds which directly transverse or which induce fatigue in high-stress members, and spot, tack, or intermittent welds for strength are not acceptable. Deficiencies in workmanship shall not be covered up by using filler materials. Fillet welds shall be provided, when necessary, to reduce stress concentrations. The surfaces to be welded shall be free from scale, paint, grease, and other foreign matter. When multiple layers of weld metal are required, each layer, or each layer after the third pass with multipass welding process, shall be thoroughly cleaned before depositing another layer. All welded parts and assemblies shall be free from cracks and other imperfections that may reduce the effectiveness of the part or assembly. After welding, all weldments shall be free from slag (except mill scale), flux, weld spatter, and other factors which affect the quality of the weld. Except for vertical welding of the electroslag process, the work shall be positioned for flat welding whenever practicable. No undercuts or overlays will be permitted on any weldment. Butt-type joints shall be alined within 20, 10, 7.5, and 5 percent of the thickness of the equal or thinnest members involved for metal thicknesses of 1/32, 1/4, 1/2, and 1 inch minimums, respectively. Where different metals are welded to each other, filler metal having strength of the lower strength material shall be used. When the same metals, or metals having similar properties are used, the deposited filler metal shall develop the full design strength of the particular weld joint or weldment. Mechanical properties of all welds shall be such that welds will transmit stress without failure or permanent deformation when subjected to any test specified in this specification. Weld joints subject to stresses exceeding 50,000 psi shall be inspected for soundness. Prior to production fabrication of any weldment of these metals, the contractor shall determine the inspection medium (magnetic particle, radiographic, dye penetrant, ultrasonic, gamma ray, or eddy current) necessary to determine the soundness of production welds.

3.34.2.3 Aluminum welding. Aluminum welding shall be accomplished by the gas-metal-arc or the gas-tungsten-arc method or any other method that will provide equivalent mechanical properties of the filler metal to base metal combinations. When machines are available, all aluminum welds shall be made by automatic machines. All welds shall be made at a uniform rate of speed. Complete and uniform penetration and fusion of the metals shall be obtained on all welds. Aluminum welds may be ground, filed, wire brushed or chipped; hammered welds shall not be acceptable. Preheating for welding is permissible, providing the temperature does not exceed 400° F for a total time of 30 minutes. Aluminum weld processes requiring the use of welding flux shall not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

3.34.3 Brazing. Brazing of aluminum and aluminum alloys shall be performed in accordance with nationally recognized and accepted recommended procedures and standards of the aluminum manufacturing industry.

3.34.4 Bolted connections. Boltholes shall be accurately punched or drilled and shall have burrs removed. Integral or separate washers or lockwashers shall be provided. Self-locking nuts are acceptable in lieu of standard nuts and lockwashers. All capscrews, bolts, and nuts shall be in accordance with SAE and IFI Standards and recommendations and shall be tight and correctly torqued.

3.34.4.1 Critical bolted connections. Critical load carrying bolted connections shall, in addition, be as specified herein. Boltholes shall be reamed as required. All critical junctures and joints shall be fastened with high strength bolts with washers and nuts, hex or 12-point type. Bolts shall conform to IFI Standards 103 and 111, minimum Grade 8. Washers shall be of the flat hardened steel type, sized to spread the clamping loads such that brinnelling or embedding in the component members shall be prevented. Mating nuts shall be of the prevailing torque type in accordance with IFI Standards 100, 101, 107, and 108, minimum Class C. Fasteners shall be lubricated prior to wrenching. Installation tools shall be proofed at set intervals to insure constant and correct torque performance. Fastener installation accuracy shall be achieved by the turn-of-the-fastener method.

3.34.5 Riveted connections. Riveted connections shall be in accordance with SAE J492. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.34.6 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or other defects which might render the casting unsound for its intended use.

3.34.7 Forgings. Forged components shall be in accordance with the *Forging Industry Association Handbook*.

3.34.8 Machine work. All machined parts shall be manufactured to gage through the use of correct jigs and fixtures and shall permit interchangeability of these parts.

3.34.9 Cleaniness of hydraulic and pneumatic systems. The hydraulic and pneumatic systems of the unit bodies shall be purged to remove any foreign material after the appropriate assembly phase of the van body is completed. The sequence of purging shall be first through the lines and then second through both the lines and the components.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Acceptability criteria. Units which conform to all requirements in Sections 3 and 5 of this specification and pass all applicable examinations and tests in Section 4 of this specification will be considered acceptable by the Government.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. Prior to testing, the preproduction unit shall be examined for defects as specified in 4.6.1. Presence of one or more defects shall be cause for rejection of the preproduction unit.

4.3.2 Tests. The preproduction unit shall be subjected to the tests marked "X" in Column 1 of Table VII. Failure of any test shall be cause for rejection of the preproduction unit.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production units will be selected at random by the Government from units being produced from production tooling, and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in Column 2 of Table VII to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

Government. Acceptance of an initial production unit shall not exclude the remaining units from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial pro duction test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production unit to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production unit(s) and shall be cause for refusal by the Government to continue acceptance of production units until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on units previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all units accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each unit shall be examined for the defects as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each unit shall be subjected to the tests marked "X" in Column 3 of Table VII, except that (1) the duration of the endurance test shall be 48 hours in lieu of 250 hours, (2) the mobility test shall be 10 miles distance over highways and 5 miles distance over cross-country terrain, in lieu of the total 300 miles, and (3) tests 4.6.2.7.1 (a) through (h) and 4.6.2.7.2 need not be conducted. Failure of any test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. The unit shall be examined as specified herein for the following defects:

101. Weight or dimensions not as specified.
102. Material not as specified.
103. Corrosion-resisting treatment not as specified.
104. Presence of nonpermissible galvanic couples.
105. Safety provisions or equipment missing or not as specified.
106. Human factors engineering not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T04:06Z
Check the source to verify that this is the currentversion before use.

107. Carbon-dioxide system: Components missing or not as specified.
108. Refrigeration system not as specified.
109. Water pump not as specified.
110. Semitrailer, Type I components missing or not as specified.
111. Chock blocks and brackets not as specified.
112. Vehicular electrical system not as specified.
113. Unit body construction not as specified or components missing.
114. Atmosphere conditioning provisions. Components missing or incorrectly installed.
115. Unit electrical systems and equipment not as specified or components missing.
116. Unit components and equipment not as specified for ISO container conversion unit Type II.
117. Piping, fittings, valves, and connections missing or not as specified.
118. Accessories not as specified.
119. Hose couplings not as specified.
120. Transfer and equalizer hose not as specified.
121. Fastening devices inoperable or not as specified.
122. Lubrication and lubrication devices not as specified.
123. Fungus and moisture resistance not as specified.
124. Fire extinguisher not present or not mounted as specified.
125. Identification and marking not as specified.
126. Treatment and painting not as specified.
127. Marking not as specified.
128. Workmanship not as specified.

4.6.2 Tests.

4.6.2.1 Test conditions. Prior to testing, the unit shall be serviced with military oils and greases. Oils shall be those designated for use in the ambient temperature at the place of the test. The unit shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared or submitted by the contractor prior to test. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. Unless otherwise specified herein, tests shall be conducted with all unit components and accessories installed and functioning normally. The capacity shall be measured by weight of product converted as specified in 3.11.1. Tests shall be in accordance with Table VII. It is not required that tests be performed in the order listed, except that testing of the pressure vessels and relief valves shall precede all other tests. The movement to a new location test, transportability, and setting-up after movement shall be performed at the midpoint of the endurance test. The rail impact test may be conducted at any

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

time during the test, however at the conclusion of the last test the unit shall be put into operation for not less than 8 hours at capacity production, as specified in 4.6.2.15.

Table VII. Test Schedule

Preproduction	Initial Production	Individual	Inspection Comparison	Tests	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6	7
X	X	X	X	Test conditions.	4.6.2.1	Section 3
X	-	X	-	Preassembly.	4.6.2.2	3.1, 3.12.3
X	-	-	-	Rupture discs.	4.6.2.2.1	3.12.4
X	X	X	X	Safety valves.	4.6.2.2.2	3.12.4
X	-	-	-	Sound level.	4.6.2.3	3.8.1
X	X	-	X	Maintainability.	4.6.2.4	3.10
-	X	-	X	Reliability.	4.6.2.5	3.9
X	X	X	X	Endurance.	4.6.2.6	3.11.1.2
X	-	-	-	Movement to new location.	4.6.2.6.1	3.11.1.3 & 3.11.1.4
X	-	X	-	Road.	4.6.2.7.1	3.7, 3.11, & 3.11.2 thru 3.11.2.5
X	-	X	-	Brake.	4.6.2.7.2	3.11.2.6
X	X	-	-	Set up after movement.	4.6.2.8	3.11.1.4
X	X	-	-	Lifting attachments.	4.6.2.9	3.17.2 & 3.22
X	X	-	-	Tiedown attachment.	4.6.2.10	3.17.2 & 3.23
X	X	-	-	Rail hump.	4.6.2.11	3.11.2.7
X	-	-	-	Standard air leakage.	4.6.2.12.1	3.16.4
X	-	-	-	High temperature.	4.6.2.12.2	3.5.1 & 3.5.2
X	-	-	-	Low temperature.	4.6.2.12.3	3.5.2
X	-	-	-	Rain.	4.6.2.12.4	3.5.3
X	X	-	-	Electromagnetic interference.	4.6.2.13	3.18
X	-	X	-	Human factors.	4.6.2.14	3.8
X	X	X	X	Operation.	4.6.2.15	Section 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

4.6.2.2 Preassembly. Prior to assembly, pressure vessels, components and accessories subject to internal pressures shall be tested in accordance with ASME Boiler and Pressure Vessel Code, Section VIII. Presence of the ASME official Code U symbol stamped or marked on a vessel and the furnishing of a copy of the ASME data sheet (Form U-1) may be accepted as evidence that the vessel conforms to the ASME Rules for Construction of Unfired Pressure Vessels. Inability to pass the test or lack of the official stamp or mark shall constitute failure of this test.

4.6.2.2.1 Rupture discs. A sample rupture disc of each type used shall be tested as specified in the ASME Boiler and Pressure Vessel Code, Section VIII. Inability of the rupture disc to meet the requirements of the ASME Boiler and Pressure Vessel Code shall constitute failure of this test.

4.6.2.2.2 Safety valves. Each safety valve shall be tested in accordance with the ASME Boiler and Pressure Vessel Code, Section VIII. Inability of the safety valve to meet the requirements of the ASME Boiler and Pressure Vessel Code shall constitute failure of this test.

4.6.2.3 Sound level measurement.

4.6.2.3.1 Operator's station. Measure the sound level at the press operator's station in accordance with MIL-STD-1474 for steady-state noise while producing dry ice at maximum capacity. Nonconformance to 3.8.1 shall constitute failure of this test.

4.6.2.3.2 Required test at 50 feet. Measure the sound level at 50 feet in accordance with SAE J952, except the meter response speed shall be slow. Note, this requires that all directions be considered for a stationary test or the loudest side of a drive by test, whichever is applicable. Nonconformance to 3.8.1 shall constitute failure of the test.

4.6.2.4 Maintainability. While performing the endurance test, two 1/2-hour periods each 50 hours will be permitted for normal scheduled adjustment and servicing as specified and all periodic preventive maintenance and servicing operations shall be performed to determine ease of accomplishment and accessibility of parts and assemblies. These operations shall be conducted as part of the normal preventive maintenance and servicing operations performed in accomplishing the endurance and performance testing specified herein. Nonconformance to 3.10 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

4.6.2.5 Reliability. Using the MTBF specified in 3.9, the initial production unit shall be tested as specified in 4.6.2.6 with "accept" and "reject" criteria in accordance with Figure 5. A mission is considered 12 hours of operating time. Continue testing until either an "accept" or "reject" decision is reached. A failure is defined as any malfunction which cannot be corrected within 30 minutes (all waiting periods are excluded) by adjustment, repair, or replacement action using controls, and on-equipment tools or parts available to the parent unit organization when deployed, and which causes or may cause:

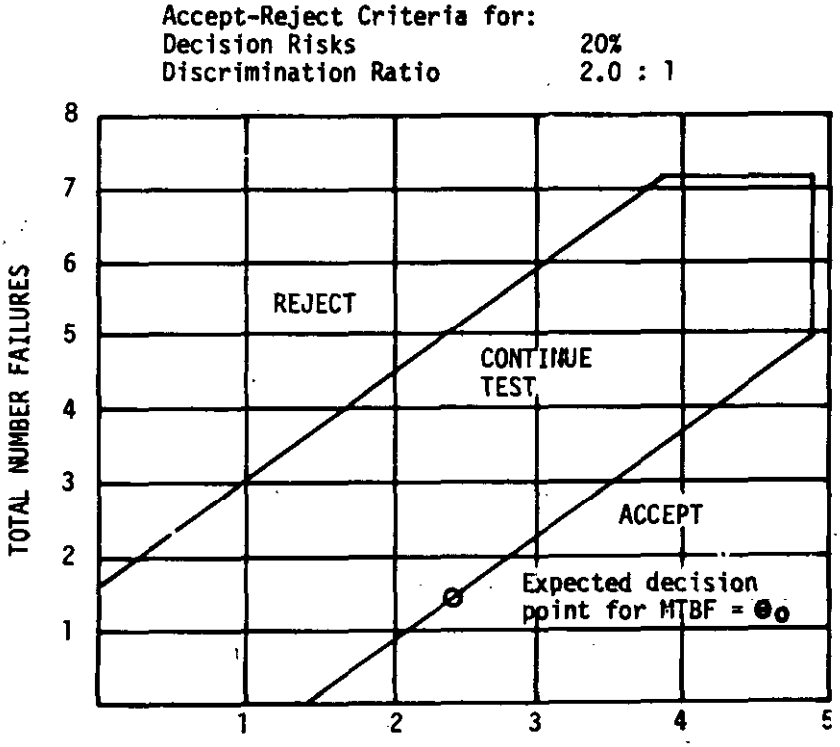
(a) Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
(b) Damage to the unit by continued operation.
(c) Safety hazards to personnel.

Note. Prior to starting the test the unit shall be run-in for not less than 20 hours.

4.6.2.5.1 Maintenance support. During testing as specified in 4.6.2.6, maintenance support elements such as manuals (when furnished as required by the contract), lubrication charts, repair parts, and special tools shall be the support elements shall constitute failure of this test and shall require revision of the document(s). Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the MTBF.

4.6.2.6 Endurance. Operate the conversion unit for 250 hours in 8 hour shifts at the normal feed rate of 300 pounds of liquid carbon dioxide per hour at 300 psig and 0° F. Maintain a complete log of operating conditions. Readings shall include all essential data, including operating temperature and pressure and capacity and shall be taken every 60 minutes. Upon completion of 125 hours of operation shut down the unit and prepare the unit for movement as specified in 4.6.2.6.1. Then perform the transportability test specified in 4.6.2.7. After successful completion of the transportability test and arrival at a new location perform the setting up test as specified in 4.6.2.8. After successful completion of the setting up test operate the unit as specified herein for the remaining 125 hours. Inability to produce 260 pounds of dry ice per hour in the specified dimensional form or purity, failure of any component, nonconformance to 3.11.1 through 3.11.1.4, or inability to set up the unit and begin production of dry ice within 2 hours shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:06Z
Check the source to verify that this is the currentversion before use.



Number of Failures	Total Test Time* Reject (Equal or less)	Accept (Equal or more)
0	N/A	1.40
1	N/A	2.09
2	.35	2.79
3	1.04	3.48
4	1.73	4.17
5	2.43	4.87
6	3.12	4.87
7	3.81	4.87
8	4.87	N/A

*Total test time is total unit hours of "equipment on" time and is expressed in multiples of the specified MTBF.

FIGURE 5. Accept-reject criteria.

X-3490

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:06Z
Check the source to verify that this is the current version before use.

4.6.2.6.1 Movement to new location. The unit shall be started and operated until it is producing capacity of product, then shut down and prepared by two trained operators for on- and off-road movement. The time required to ready a Type I unit and couple it to the truck-tractor or the time required to ready a Type II unit and load and secure it on a low bed trailer or ISO transporter shall be recorded. Nonconformance to 3.11.1.3 and 3.11.1.4 shall constitute failure of this test.

4.6.2.7 Transportability.

4.6.2.7.1 Road. The Type I unit shall be tested while coupled to the specified truck-tractor. The Type II unit shall be loaded on the specified low-bed semitrailer or on an ISO transporter semitrailer and then coupled to the specified truck-tractor. Semitrailers, which are not furnished under the contract and their payloads of Type II conversion units, shall be subjected to the road test in Table VIII, but shall be exempt from running the tests herein of (a) through (h) and the brake tests in 4.6.2.7.2. The unit shall be towed as specified in Table VIII for a total distance of not less than 300 miles.

Table VIII. Road Test

Course	Mileage and Speeds
Primary highway	200 mi. at variable speeds up to 60 mph
Secondary Roads	50 mi. at variable speeds up to 30 mph
Off-Road	50 mi. at variable speeds up to 10 mph

During the road test for Type I units with a semitrailer, perform and record the actual results of the following:

(a) Tow the unit on side slopes up to 10 percent, having the curb side down hill and, again, having the road side down hill. Make full-throttle starts and panic stops on the side slope.
(b) Tow the unit over 8-inch-high obstacles such that all combinations of the wheels have passed over the obstacles. Observe tire-to-surface contact, evidence of binding or interference, and, after returning the suspension to normal, measure the diagonals and parallels for evidence of misalinement. Questionable tire-to-surface contact shall be retested with single sheets of newspaper under the tires. Ability to remove the newspaper from under the tire shall be evidence of insufficient tire-to-surface contact.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

(c) Tow the unit over highway type V-ditches (US Bureau of Public Roads Type) not less than 36 inches wide and 18 inches deep. Negotiate the ditch both perpendicular and at an angle of 45 degrees to the direction of travel until at least 10 V-ditch crossings have been completed. Observe the semitrailer as first the forward wheels and then the rearward wheels lose contact with the ground.

(d) Measure and record the deviation or off tracking of the unit with respect to the prime mover, when the unit is towed straight ahead on a paved surface.

(e) Jackknife the unit at angles up to 90 degrees from straight ahead both to the left and to the right.

(f) Turn the prime mover, towing the unit, in a rolling, sharpest-turning circle for two complete turns and repeat in the opposite direction. Observe and inspect the suspension.

(g) Park the tractor-trailer combination on a paved surface. Lower the landing gear to a height of 4-inches from the pavement. Uncouple the brake lines, setting the brakes, and unlock the tractor fifth wheel jaws. With the tractor in low-low or in 2nd-low, quickly engage the engine to the drive train and immediately accelerate the tractor out from under the front of the unit to rapidly remove all support at the front of the trailer, allowing the landing gear pad to strike the paved surface from the free fall height. Repeat landing gear impacts for a total of two free falls. Inspect landing gear and trailer. Operate the landing gear from fully retracted to full raised not less than five times.

(h) Turn the semitrailer lights on, and immerse the semitrailer into water not less than 20 inches deep for not less than 2 minutes. Stop the unit using the unit brakes and turn off the lights for not less than 10 minutes. Turn on the lights. Tow the unit out of the water and make a brake stop using the unit brakes and observe unit braking for water fade, grabbing, and chatter.

Failure to complete the test, inability to operate and function as specified herein, or nonconformance to 3.11.2 through 3.11.2.5 shall constitute failure of this test.

4.6.2.7.2 Brake. Test the units with a semitrailer in accordance with the DOT Federal Motor Vehicle Safety Standards, Motor Carrier Safety Regulations, the California Vehicle Code and to the requirements specified. Tests shall be conducted after the unit has been prepared for movement (see 3.11.1.3). Test the unit brake system as specified in TTMA RP #12 and, when applicable, RP #44. Failure to complete the test, inability to function and operate as specified, nonconformance to DOT Regulations and Standards, the California Code, or 3.11.2.6 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:06Z
Check the source to verify that this is the current version before use.

4.6.2.8 Setting-up after movement. For units with a semitrailer, on arrival at a new location, the truck-tractor shall be uncoupled and the unit prepared for operation by two trained operators. The time consumed between uncoupling from the truck-tractor and the production of dry ice shall be recorded. For the Type II units without a furnished semitrailer on arriving at a new level, hard surfaced location, the unit shall be unloaded from the low-bed trailer or transporter. The time consumed between beginning the unloading and the production of dry ice shall be recorded. Inability to set up the units and begin production of dry ice by two trained operators within 2 hours shall constitute failure of this test.

4.6.2.9 Lifting attachments. Test the unit's lifting attachments as follows:

(a) Lift the unit and hold in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the unit. Determine the forward angles of application of each attachment. Measure the clearance between each sling and the unit. Measure eye openings and clearance dimensions of the attachment eyes.
(b) Restrain the unit by anchoring the main frame and subject each attachment to a pull of 2-1/2 times the load it carried when initially suspended. This pull shall be applied in the direction as determined in (a) above. An alternate method of test is to lift the unit as in (a) above and add weights to the main frame until the force is 2-1/2 times the initial force on each attachment. Hold each load for 90 seconds.

Weld failure, or permanent deformation of the unit or its lifting attachments, clearance of less than 1-inch between the slings and the unit except where a structure (such as an overhead guard) attached to the unit has sufficient strength to withstand contact with the sling and incurs no permanent deformation, slings converging more than 24 feet above the lowest extremity of the unit or failure of dimensions to conform to MIL-STD-209, shall constitute failure of this test.

4.6.2.10 Tiedown attachments. With the unit anchored by means other than the attachment being tested, apply the proportion of the static loads as specified in MIL-STD-209 to each attachment. Measure dimensions of the tiedowns. Breakage or deformation of the tiedown such that it can no longer withstand the required static loads or failure to conform to dimensions in MIL-STD-209 shall constitute failure of this test. Local deformation in the tiedown and nearby vicinity which does not effect the useability of the attachment does not constitute failure of the test.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:06Z
Check the source to verify that this is the currentversion before use.

4.6.2.11 Railroad humping. Anchor the unit, both the Type I and Type II, on a domestic service standard railway flatcar using the tiedown attachments and any blocking required to prevent movement of the unit. Hump the car, with the unit tied down, at a speed of not less than 8 mph against five empty boxcars with their brakes set. The unit shall face in the direction of the five cars. Repeat the procedure with the rear of the unit in the direction of the five cars. The unit may be reversed by either reloading it in the proper direction on the flat car or by reversing the direction of the flatcar. Once the test has begun, there will be no readjustment of load, nor any reconditioning of the blocks, chocking material or tiedowns. After each impact, the unit and tiedowns shall be inspected to determine permanent deformation to components of the unit or other evidence of possible failure. After completion of the railroad impact test, an operational check-out of the unit will be conducted. Weld failure, or permanent deformation to the components of the unit or its tiedown attachments, stopping of a test due to impeding failure and possible damage to the unit, or failure of the unit to operate after the test shall constitute failure of the test.

4.6.2.12 Environmental.

4.6.2.12.1 Standard air leakage rating test. The unit body shall be tested or certified to determine the heat transmission rating and shall be tested to determine the air leakage rate. All certification and tests shall be in accordance with TTMA RP No. 38 or ISO/ANSI methods. Absence of a TTMA thermal rating plate on each unit or nonconformance to 3.16.4, Table IV, (a), shall constitute failure of this test.

4.6.2.12.2 High temperature. The unit shall be stored in an ambient temperature of plus 155° F or plus 68.3° C for 4 hours. The unit shall be examined for damage. The unit shall then be stabilized at plus 125° F or plus 51.67° C. While at this temperature and in an equal ambient temperature, the unit shall be operated for 8 hours. Damage to or permanent deformation of any part of the unit, or a production capacity and quality of product less than that specified in 3.5.1 and 3.11.1 shall constitute failure of this test.

4.6.2.12.3 Low temperature. Subject the unit to storage in an ambient temperature of minus 50° F or minus 45.56° C without benefit of solar radiation until all parts have stabilized at minus 50° F or minus 45.56° C. Maintain the unit at this temperature for not less than 6 hours, after which the unit shall be visually examined for damage or deformation of components and instruments. The unit shall then be stabilized at minus 25° F or minus 3.89° C and operated at the specified production rate for 8 hours. At the conclusion of the performance run the unit shall be examined. Evidence of

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:06Z
Check the source to verify that this is the current version before use.

leakage, malfunction of controls, cracks, permanent deformation, production capacity less than that specified in 3.11.1 or inability of the space heaters to operate as specified in 3.14.7 shall constitute failure of this test.

4.6.2.12.4 Rain. The complete unit with all machinery and mounted equipment shall be subjected to a simulated rainstorm, while prepared for storage and during normal operation. The apparatus for producing the spray shall emit the water in the form of droplets rather than as a fine mist at the rate per minute and in the direction specified in 3.5.3. Water leakage through the roof and sides of the unit body when the unit is prepared for storage, inability of the unit to operate as specified, or exposure of operator personnel to the precipitation, shall constitute failure of this test.

4.6.2.13 Electromagnetic interference. The unit shall be subjected to the tests specified in MIL-STD-461 for electrical, electro/mechanical equipment. The contractor shall furnish the contracting officer with the test report for evaluation, within the time frame specified (see 6.2). Disapproval of the report shall constitute failure of this test.

4.6.2.14 Human factors. The unit shall be evaluated to determine the human factors criteria specified in 3.8. Nonconformance to 3.8 shall constitute failure of this test.

4.6.2.15 Operational test. After all other tests have been completed, a final 8 hour operational test shall be performed. The conversion units shall be operated at the normal input feed rate of 300 pounds of liquid carbon dioxide per hour at 300 psig and 0° F, or minus 17.78° C and 260 pounds of dry ice per hour output. A complete log of operating conditions shall be maintained. Readings shall include all essential data, including operating temperature, pressure, and capacity. These readings shall be taken and recorded once each 60 minutes of operation. Upon the satisfactory completion of the 8 hours of operation, the units shall be shutdown. Inability of the conversion unit to complete the 8 hours of operation constitutes failure of the test.

4.7 Inspection comparison. When initial production inspection is specified (see 3.3), the Government may select units at any time during the contract production period and subject these units to the examination specified in 4.6.1 and to the tests marked "X" in Column 4 of Table VII to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Units will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:06Z
Check the source to verify that this is the currentversion before use.

initial production units. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison unit to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison units and shall be cause for refusal by the Government to continue acceptance of production units until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on units previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all units accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. Examine the preproduction pack for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

129.	Materials, methods, or containers not as specified for Level A.	Each incorrect material, method, or container shall be considered one defect.
130.	Disassembly and matchmarking not as specified for Level A.	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

131. Unpainted exterior metal surfaces not coated with preservative for Level A.
132. Semitrailer chassis and body not preserved in accordance with the referenced document for Level A.
133. Carbon dioxide system not drained and dried, valves not closed, and openings not sealed as specified for Level A.
134. Electric motors not preserved in accordance with the referenced document for Level A.
135. Carbon dioxide compressor not preserved in accordance with the referenced document and exception for Level A or C.
136. Refrigeration system not drained and dried, valves not closed, and openings not sealed as specified for Level A.
137. Water pump and water system not drained and valves not left open for Level A.
138. Dry ice press not preserved as specified for Level A.
139. Electrical extension cables and hoses not placed and secured in the storage area provided for Level A.
140. Scales not locked, blocked, cushioned, and preserved as specified for Level A.
141. Bandsaw components and accessories not preserved in accordance with the referenced document for Level A.
142. Drivebelts not released from tension and unpainted pulleys not coated with primer as specified for Level A.
143. Removed gages and instruments not preserved as specified for Level A.
144. Fire extinguishers not placed and secured in the mounting brackets for Level A.
145. Maintenance tools not preserved in accordance with the referenced document for Level A.
146. Consolidation not as specified for Level A.
147. Containers and movable parts of the conversion unit components not blocked and braced as specified for Level A.
148. Marking missing, illegible, incorrect, or incomplete for Level A or C.

5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination within the time frame specified (see 6.2) to prove, prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.9). The preproduction pack may be accomplished utilizing either the preproduction unit or a

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

production conversion unit. If the preproduction model conversion unit is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government when requested by the Government to facilitate comparison between the preproduction conversion unit and production conversion units.

5.2 Preservation. Preservation shall be Level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly and matchmarking.

5.2.1.1.1 Disassembly. Disassembly of the components, parts, and assemblies of the conversion unit shall be the minimum necessary to safeguard parts known to be subject to damage or loss. Removed bolts, nuts, pins, and washers shall be placed in one of the mating parts and secured to prevent their loss.

5.2.1.1.2 Matchmarking. Removed components and parts shall be matchmarked to facilitate reassembly. Matchmarking information shall be on cloth shipping tags conforming to UU-T-81, Type A, attached to mating parts. Marking on tags shall be with waterproof ink.

5.2.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116, except preservative lubricating oil specified herein shall conform to MIL-L-21260, type and grade as specified.

5.2.1.3 Unprotected surfaces. Unpainted and threaded exterior surfaces of items, components, parts, and assemblies subject to corrosion shall be coated with Type P-1 preservative.

5.2.1.4 Semitrailer chassis and body. The components of the semitrailer chassis and body shall be preserved in accordance with the Level A (mobile) requirements of MIL-STD-281.

5.2.1.4.1 Heater. The fuel pump, fuel tank, and the fuel system shall be drained and the interior surfaces, including those of filler caps and burner, shall be coated with Type I, Grade 30, preservative lubricating oil. Excess preservative shall be drained, and the filler caps and drain plugs shall be reinstalled. Unprotected threaded and unpainted metal surfaces of components subject to corrosion shall be coated with Type P-1 preservative.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

5.2.1.5 Carbon dioxide system. The carbon dioxide system shall be drained. Interior surfaces of the system shall be dried with dry compressed air. Valves shall be closed, and openings into the system shall be sealed with plastic caps or plugs with tap conforming to PPP-T-60, Type IV.

5.2.1.5.1 Electric motors. Electric motors shall be preserved in accordance with the Level A requirements of MIL-E-16298 using the alternate method as specified for assembled machines.

5.2.1.5.2 Carbon-dioxide compressor. The carbon-dioxide compressor shall be preserved in accordance with the Level A requirements of MIL-C-3600, except no preservatives shall be applied to interior surfaces of the compressor and components that will contact carbon dioxide.

5.2.1.5.3 Refrigeration system. The refrigeration system shall be preserved in accordance with MIL-R-3593, Level A.

5.2.1.5.4 Water pump. The water pump and water system shall be drained, and the drains shall be left open. Exterior unpainted surfaces of the pump shall be coated with Type P-1 preservative.

5.2.1.5.5 Dry-ice press. The hydraulic fluid supply tank shall be drained and refilled to the operating level with hydraulic fluid required by the technical manual supplied with the conversion unit, and the press shall be actuated to insure coating of all interior surfaces of the pumps, piping, and valves with the hydraulic fluid. The pressure platens shall be coated with Type P-6 preservative, and the surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, Type II, Grade A, Class 2, secured in place with tape conforming to PPP-T-60, Type IV. Control levers shall be secured to prevent movement.

5.2.1.6 Electrical extension cables. The electrical extension cables and hoses shall be placed and secured in the storage area provided.

5.2.1.7 Ground rods and ground cables. The driving bolts shall be coated with Type P-1 preservative. Each ground rod assembly with driving bolt and cables shall be preserved together in accordance with MIL-P-116, Method III, using a close-fitting box conforming to PPP-B-636, W6c.

5.2.1.8 Scales. The scales shall be locked, blocked, and cushioned to prevent damage by vibration. Unpainted exterior machined surfaces shall be coated with Type P-6 preservative and the coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, Type II, Grade A, Class 2 secured in place with tape conforming to PPP-T-60, Type IV.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:06Z
Check the source to verify that this is the current version before use.

5.2.1.9 Bandsaw. The bandsaw, components, and accessories shall be preserved in accordance with the Level A requirements of MIL-M-18058.

5.2.1.10 Drive belts and pulleys. Drive belts shall be released from tension. Unpainted pulley grooves shall be coated with primer conforming to TT-P-664.

5.2.1.11 Gages and instruments. Gages and instruments removed from the conversion unit shall be individually preserved in accordance with MIL-P-116, Method IC-2. The containers shall be boxes conforming to PPP-B-566, style, type, and class optional or the contractor's commercial container.

5.2.1.12 Fire extinguishers. Unpainted metal surfaces of the fire extinguishers subject to corrosion shall be coated with Type P-1 preservative. The extinguishers shall be placed in the mounting brackets and secured.

5.2.1.13 Maintenance tools. Maintenance tools shall be preserved in accordance with the Level A requirements of PPP-P-40.

5.2.1.14 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve the repair parts.

5.2.1.15 Technical publications. Technical publications for each conversion unit shall be preserved together in accordance with MIL-P-116, Method IC-3 or IC-1.

5.2.1.16 Consolidation. Components and assemblies shall be placed either in the compartments provided or in the toolbox. Components for which no compartment is furnished or those that will not fit in the toolbox shall be consolidated in close-fitting boxes conforming to PPP-B-601, Domestic Type, style optional. Size of the boxes shall be determined by the space available for placing and securing in the unit body.

5.2.2 Level C. The conversion units, components, parts, repair parts, tools, and publications shall be preserved to afford protection against deterioration and damage from the contractor to the initial destination, except no preservatives shall be applied to surfaces of the carbon-dioxide compressor and components that will contact carbon dioxide.

5.3 Packing (see 6.10). Each complete semitrailer-mounted or ISO container conversion unit with all accessories, repair parts, tools, and publications enclosed within the body shall be prepared for shipment mobile in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

National Motor Freight Classification rules. Components and equipment, installed within the body, and any other movable equipment or items, shall be cushioned, blocked, braced, and anchored to prevent damage to the equipment or to the body. The doors shall be secured to prevent accidental opening and in a manner to prevent pilferage of contents.

5.4 Marking. In addition to any special marking required, components, boxes, semitrailers and ISO containers shall be marked in accordance with MIL-STD-129 (see 6.10).

6. NOTES

6.1 Intended use. The carbon-dioxide conversion unit is intended for the manufacture of dry ice (solid carbon dioxide) from liquid carbon dioxide.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of unit required (see 1.2 and 3.1).
(c) Time frame required for submission of preproduction model and number of preproduction units required (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(e) When initial production testing is required and number of initial production units required when applicable (see 3.3).
(f) When pellet form of dry ice is required in lieu of the block form (see 3.11.1 and 3.13.8).
(g) Method of thawing if other than as specified (see 3.13.2).
(h) When unused floor space is to be utilized as specified in 3.13.8.
(i) When a power cable other than as specified is required (see 3.15.1).
(j) When an auxiliary power cable is required (see 3.15.2).
(k) When a 24 volt military vehicle electrical system is required for the 12-volt semitrailers to be compatible with M Series trucks (see 3.16.3).
(l) When the conversion unit shall be mounted all in one each, Size 1A, 12 000 mm (40 ft) ISO container (see 3.17).
(m) When the overall exterior and interior maximum width of the container conforming to the ISO width limit of 102 inches (2.59 meters) and the inside width increased to 95 inches minimum, is required (see 3.17.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

(n) When a container transporter semitrailer is required for each unit and the semitrailer characteristics are required (see 3.17.3).
(o) When colors other than as specified are required (see 3.29).
(p) When Government furnished property is required (see 3.32 and 6.7).
(q) Time frame required for submittal of electromagnetic interference test report (see 4.6.2.13).
(r) Time frame required for submittal of preproduction pack (see 5.1).
(s) Level of preservation required (see 5.2).

6.3 Preproduction model. Any changes or deviations of production units from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish units conforming to this specification (see 3.2).

6.4 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the unit covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91 (see 3.19.1).

6.5 Registration numbers. The contracting officer should furnish a list of registration numbers to be applied to the units (see 3.33).

6.6 Vehicle weight classification forms. The contracting officer should arrange to furnish the contractor with the applicable Vehicle Weight Classification Forms and instructions for completion. Further data may be obtained by contacting: Commander, US Army Mobility Equipment Research and Development Command, ATTN: DRDME-HM, Fort Belvoir, VA 22060 (see 3.33).

6.7 Government furnished property. The contracting officer should arrange to furnish the property specified (see 3.32 and 6.2).

6.8 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each unit.

6.9 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the units in accordance with this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

6.10 Packing levels. Levels of packing have not been included as the nature of the item is such that only mobile packing requirements are applicable. For purposes of preservation/packing level markings, the level of packing shall be considered Level A and shall be marked A/A, or C/A as applicable.

6.11 Provisioning. The contracting officer should include provisioning requirements for repair and maintenance tools as necessary, including special tools and instructions on shipment of the converter.

6.12 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activity:
DLA - GS

Project 3655-A085

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-C-52025C(ME) Conversion Units, Carbon-Dioxide, 260 Pounds Per Hour

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Mobility Equipment Research and
Development Command
ATTN: DRDME-DS
Ft. Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.